SUPERLINEAR WIRE LOWER BOUNDS FOR GATES OF FAN-IN $N^{1-\delta}$

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Abstract. For every $\delta > 0$, a language in E^{NP} needs more than $c(\delta)N\sqrt{\log N}$ wires in circuits of any depth on N inputs whose gates are arbitrary Boolean functions of at most $N^{1-\delta}$ inputs, or threshold, exact, modular or symmetric gates of any fan-in with weights of at most $N^{1-\delta}$ bits. The proof extends the tensor-network contraction theorem of OpenAI from degree $s^{9/10}$ to degree $s^{1-\delta}$ and counts the satisfying inputs of such circuits in time $2^{O(s/\sqrt{\log s})}$. The P versus NP problem remains open, and we show where the method stops.

Keywords: P versus NP; circuit lower bounds; tensor networks; threshold circuits; exponential time hypothesis.
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1. Introduction

1.1. **The problem.** The class P consists of the languages decided by a deterministic Turing machine in time polynomial in the input length, and NP of those decided by a nondeterministic one in polynomial time, equivalently those whose members have certificates of polynomial

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length that can be checked in polynomial time. The question whether $\mathsf{P} = \mathsf{NP}$ was isolated by Cook [Coo71] and Levin [Lev73], who showed that satisfiability of propositional formulas is NP-complete; Karp [Kar72] then showed that twenty-one combinatorial problems share this property. The question had been anticipated by Edmonds [Edm65], who proposed polynomial running time as the mark of an efficient algorithm, and by Gödel in a letter of 1956 to von Neumann, discussed by Sipser [Sip92]. It is one of the seven Millennium Prize Problems of the Clay Mathematics Institute [Clay], and the surveys of Sipser [Sip92], Fortnow [For09], Aaronson [Aar16] and Wigderson [Wig19] describe its place in mathematics.[1]

No proof that $\mathsf{P} \neq \mathsf{NP}$, or that $\mathsf{P} = \mathsf{NP}$, is known, and this paper does not give one. Its subject is the strongest kind of statement that is currently within reach: lower bounds for explicit languages against restricted models of computation. The most direct route to $\mathsf{P} \neq \mathsf{NP}$ is to show that some language in NP has no Boolean circuits of polynomial size. Shannon [Sha49] showed that almost every Boolean function of n inputs needs at least $2^n/(2n)$ gates of fan-in two, yet for languages in P, and even in NP, the best lower bound known is $3.1n - o(n)$ [LY22], linear in the input length n.[2] Climbing to E^{NP}, the class of languages decided in time $2^{O(n)}$ with an oracle for NP, does not help by itself: before the preprint [OAI26] discussed below, no superlinear lower bound was known for E^{NP} even for circuits of fan-in-two gates.

1.2. **What is proved.** In October 2026 OpenAI released a preprint [OAI26] with a deterministic algorithm that counts the satisfying assignments of a k-CNF with n variables in time $2^{O(n \log\log n/\sqrt{\log n})}$, which contradicts the exponential time hypothesis of Impagliazzo and Paturi [IP01].[3] The algorithm rests on a contraction theorem for Boolean tensor networks [OAI26, Theorem D.1], stated here as Theorem 8.3, and [OAI26, Corollary 1.4] combines it with Williams' argument [Wil13] to show that a language in E^{NP} has no circuits of $c\,n\sqrt{\log n}$ fan-in-two gates. The results below extend both from fan-in two to gates of fan-in $N^{1-\delta}$ for any fixed $\delta > 0$. Write $\mathrm{w}(C)$ for the number of wires of a circuit C, the sum of the fan-ins of its gates, and $s(C) = m + \mathrm{w}(C)$ when C has m inputs.

For integers $F, \beta \geq 1$ let $\mathcal{K}(F, \beta)$ be the circuits whose gates are of two kinds (Definition 8.1):

- *table gates* computing an arbitrary Boolean function of at most F inputs, given by its truth table;
- *weighted gates* of any fan-in f, computing $\varphi(\sum_i w_i y_i)$ for integer weights $|w_i| \leq 2^\beta$, where φ is a threshold $[v \geq \theta]$, an exact test $[v = \theta]$, a congruence $[v \equiv \theta \bmod \mu]$ with $1 \leq \mu \leq 2^\beta$, or, when every weight is one, an arbitrary function of $v \in \{0, \dots, f\}$ given by its table.

Gates of fan-in two, unbounded fan-in AND, OR and parity gates, majority and threshold gates with weights below 2^β, and MOD_μ gates are all of these kinds, in any combination and at any depth. Every gate of fan-in at most F, whatever it computes, is a table gate.

Theorem A (Counting, Theorem 9.1)**.** *For every $\delta \in (0, \frac{1}{2}]$ there is a deterministic algorithm that, given a circuit $C \in \mathcal{K}(s^{1-\delta}, s^{1-\delta})$ with $s = s(C) \geq 2$, described by L bits, computes the number of inputs on which C outputs one in time $(s+L)^{O(1)} 2^{O(s/\sqrt{\delta \log s})}$ and space $(s+L)^{O(1)}$. Since $L \leq s^{O(1)} 2^{s^{1-\delta}}$, the time is $2^{O(s/\sqrt{\delta \log s})}$.*

[1] Gasarch's third poll of researchers [Gas19], taken in 2019, found that 88% of those who answered expected $\mathsf{P} \neq \mathsf{NP}$, against 83% in 2012 and 61% in 2002.

[2] Throughout, an *explicit* function is one computed in polynomial time, or at least in exponential time with an NP oracle; a lower bound for a function defined by counting, as in Shannon's argument, does not name the function.

[3] The preprint has not been refereed; its proofs are being examined by a working group. The present paper reproduces in Appendix A the part of the argument it uses, so that every step it rests on is on the page.

Theorem B (Lower bound, Theorem 11.6)**.** *There is a language* $\mathsf{W} \in \mathsf{E}^{\mathsf{NP}}$ *such that for every* $\delta \in (0,1)$ *and infinitely many* N*, every circuit in* $\mathcal{K}(N^{1-\delta}, N^{1-\delta})$ *with* N *inputs that decides* W *on inputs of length* N *has more than* $c(\delta)N\sqrt{\log N}$ *wires, with* $c(\delta) = \sqrt{\min(\delta, 1/2)}/(5 \cdot 10^6)$*.*

In particular (Corollary 11.8), circuits of linear threshold gates with arbitrary real weights and fan-in at most $N^{1-\delta}$, of any depth, need more than $c(\delta)N\sqrt{\log N}$ wires to decide W. A fan-in-two circuit with g gates has at most $2g$ wires, so Theorem B contains [OAI26, Corollary 1.4]. Table gates of fan-in $N^{1-\delta}$ cannot be replaced by fan-in-two gates at linear cost: most functions of $k = N^{1-\delta}$ inputs need at least $2^k/(2k)$ such gates [Sha49]. A threshold gate of fan-in f with weights of β bits is computed by the usual adder circuits with $O(f\beta)$ fan-in-two gates, which for $\beta = N^{1-\delta}$ is far from linear in its f wires. Theorem A therefore works with the gates themselves rather than through a simulation.

The first version of this paper [Bha26v1] proved Theorems A and B for $\mathcal{K}(s^{4/5}, s^{2/3})$ and $\mathcal{K}(N^{3/4}, N^{1/2})$, using the contraction theorem as stated by OpenAI, for networks of degree at most $s^{9/10}$. The step to $N^{1-\delta}$ is a new form of the contraction theorem (Theorem 8.4), in which the degree and the number of two-adic levels are parameters. Its proof shows that the exponent $9/10$ enters the argument of OpenAI at one inequality, the balance of a labelling, and gives a time $2^{O(s/\sqrt{\delta \log s})}$ for degree $s^{1-\delta}$ and $2^{O(s/\sqrt{\log(s/\Delta)})}$ for degree Δ in general (Corollaries 8.5 and 8.6). For fan-in $N/(\log N)^K$ the same argument gives a lower bound $cN\sqrt{\log\log N}$ (Proposition 11.9).

The counting theorem also applies to integer programs. For a matrix A write $\nu(A)$ for its number of nonzero entries.

Theorem C (Zero-one programs, Corollary 10.1)**.** *Fix* $\delta \in (0,1)$*. Let* $A \in \mathbb{Z}^{r \times n}$ *and* $b \in \mathbb{Z}^r$*, and put* $\sigma = n + \nu(A)$*. If every entry of* A *is an integer of at most* $\sigma^{1-\delta}$ *bits, the number of* $x \in \{0,1\}^n$ *with* $Ax \le b$ *can be computed in time* $L^{O(1)}2^{O_\delta(\sigma/\sqrt{\log \sigma})}$*, and a linear objective with coefficients of the same size can be maximised over these points within the same bound.*

For sparse programs, with $\nu(A) = O(n)$, this is $2^{O(n/\sqrt{\log n})}$. Before [OAI26], the best algorithm for zero-one programs with $O(n)$ constraints ran in time $2^{(1-\epsilon)n}$ for a constant $\epsilon > 0$ depending on the ratio of constraints to variables [ILPS14]; a subexponential algorithm for sparse zero-one programs would already have refuted the exponential time hypothesis, since 3-SAT with $O(n)$ clauses is such a program.

The argument that turns Theorem A into Theorem B works for any exponent. Its general form is Theorem 11.5:

Theorem D (Transfer)**.** *Let* $0 < \delta_1 < \delta_2 < 1$ *and let* $\gamma > 0$ *be rational. If satisfiability of circuits in* $\mathcal{K}(s^{1-\delta_1}, s^{1-\delta_1})$ *can be decided in time* $L^{O(1)}2^{O(s/(\log s)^\gamma)}$*, then some language in* E^{NP} *needs more than* $cN(\log N)^\gamma$ *wires in* $\mathcal{K}(N^{1-\delta_2}, N^{1-\delta_2})$ *for infinitely many* N*.*

Theorem A supplies $\gamma = \frac{1}{2}$, and a $2^{O(s/\log s)}$ algorithm would give $cN \log N$. Superpolynomial lower bounds are a different matter. In Williams' theorem [Wil13] what has to be beaten is exhaustive search over the n inputs: deciding satisfiability of circuits of polynomial size with n inputs in time $2^n/n^{\omega(1)}$ would show $\mathsf{NEXP} \not\subseteq \mathsf{P/poly}$. The bound of Theorem A is measured in the size s, and it is below 2^n only while $s = O(n\sqrt{\log n})$. Section 12 explains why the method stops at $\sqrt{\log N}$ and at fan-in $N^{1-\delta}$, and what each further step would require; none of these steps would separate P from NP.

1.3. **Plan.** Part I surveys the problem. Section 2 gives definitions, history and equivalent forms; Section 3 the barriers that every proof must pass; Section 4 the circuit lower bounds known today; Section 5 the algorithmic method, the exponential time hypothesis and its refutation; Section 6 the other programmes; and Section 7 the repositories and the catalogue of 6,480 arXiv papers assembled for this survey. Part II contains the new results: circuits, tensor networks and the parametrised contraction theorem in Section 8, Theorem A in Section 9, zero-one programs in Section 10, Theorems B and D in Section 11, and the limits

of the method in Section 12, including an unconditional theorem (Theorem 12.4) showing that encodings of weighted gates by trees of tensors cannot handle weights much longer than their degree. Appendix A proves the parametrised contraction theorem, following the argument of OpenAI [OAI26, Appendix D.4] with every constant written out; Appendix B collects the constants and the calculations behind them; and Appendix C describes the Lean and Julia programs that check the finite statements.

Conventions. Logarithms are to base two unless written ln. $[n] = \{1, \dots, n\}$. For a proposition P, $[P]$ is 1 if P holds and 0 otherwise. For an integer or rational number $x \neq 0$, $v_2(x)$ is its two-adic valuation. Running times count bit operations on a random-access machine unless a Turing machine is named.

Part 1. The problem and the state of knowledge

2. Definitions, history and equivalent forms

2.1. **Time, nondeterminism and reductions.** Hartmanis and Stearns [HS65] measured the difficulty of a problem by the running time of the best Turing machine that decides it and proved the first hierarchy theorem: for time-constructible t, more time decides more languages. A nondeterministic machine may branch at each step and accepts when some branch accepts. The class $\mathsf{NP} = \bigcup_k \mathrm{NTIME}(n^k)$ is the set of languages $\mathcal{L}$ for which there are a polynomial p and a polynomial-time relation R with

$$x \in \mathcal{L} \iff \exists y \in \{0,1\}^{p(|x|)}\ R(x, y).$$

The string y is a *certificate*. A polynomial-time *reduction* from $\mathcal{L}$ to $\mathcal{L}'$ is a polynomial-time map g with $x \in \mathcal{L} \iff g(x) \in \mathcal{L}'$, and $\mathcal{L}'$ is NP-complete if it lies in NP and every language in NP reduces to it.

Cook [Coo71] and, independently, Levin [Lev73] proved that the satisfiability of propositional formulas is NP-complete.[4] The proof writes the computation of a nondeterministic machine on x as a table of $p(|x|)^2$ cells and expresses its correctness by clauses that each read a bounded window of the table. Karp [Kar72] reduced satisfiability to twenty-one problems of combinatorics and optimisation, including clique, vertex cover, Hamiltonian cycle, three-dimensional matching, knapsack and zero-one integer programming, and Garey and Johnson [GJ79] catalogued several hundred more. Consequently $\mathsf{P} = \mathsf{NP}$ if and only if any one of these problems has a polynomial-time algorithm.

Edmonds [Edm65] had argued, in his paper on matchings, that polynomial time is the right notion of a "good" algorithm. Gödel asked von Neumann in 1956 whether the existence of proofs of length n for a given formula can be decided in time linear or quadratic in n, which is the bounded form of the same question; the letter is translated and discussed by Sipser [Sip92].

2.2. **What is known to separate.** Diagonalisation separates classes defined by the same resource at different amounts. The deterministic hierarchy of Hartmanis and Stearns, and the nondeterministic hierarchies of Cook [Coo73], Seiferas, Fischer and Meyer [SFM78] and Žák [Zak83], show for instance $\mathrm{NTIME}(2^n) \neq \mathrm{NTIME}(2^{n/2})$; Žák's form, which separates unary languages, is the one used in Section 11. Between different resources very little is separated. Paul, Pippenger, Szemerédi and Trotter [PPST83] proved $\mathrm{DTIME}(n) \neq \mathrm{NTIME}(n)$ for multitape machines, a statement that does not extend to n^k. Hopcroft, Paul and Valiant [HPV77] showed that time t can be simulated in space $t/\log t$, and Williams [Wil26] improved this to space $\sqrt{t \log t}$. Fortnow, Lipton, van Melkebeek and Viglas [FLMV05] proved that

[4] Levin's note treats search problems ("universal sequential search problems") and gives six complete ones, among them tiling and set cover; Cook works with decision problems and Turing reductions. Karp's reductions are the many-one reductions used today.

satisfiability cannot be decided simultaneously in time n^c and space $n^{o(1)}$ for any c below the golden ratio.

For space, nondeterminism costs at most a square: Savitch [Sav70] proved that nondeterministic space s is contained in deterministic space s^2, so $\mathsf{PSPACE} = \mathsf{NPSPACE}$. No analogue is known for time; the best deterministic simulation of $\mathrm{NTIME}(t)$ takes time $2^{O(t)}$.

2.3. **Structure inside NP.** If $\mathsf{P} \neq \mathsf{NP}$, Ladner [Lad75] showed that NP contains problems that are neither in P nor NP-complete. Some natural problems were long thought to be of this kind; primality was placed in P by Agrawal, Kayal and Saxena [AKS04]. Counting is much stronger than deciding: Valiant [Val79b] proved the permanent of a zero-one matrix #P-complete although the corresponding decision problem, existence of a perfect matching, is in P, and Toda [Tod91] showed that the whole polynomial hierarchy reduces to #P. The PCP theorem of Arora and Safra [AS98] and of Arora, Lund, Motwani, Sudan and Szegedy [ALMSS98], with Dinur's combinatorial proof [Din07], shows that every language in NP has certificates that a randomised verifier checks by reading a constant number of bits; equivalently, approximating the maximum fraction of satisfiable clauses of a 3-CNF within some constant factor is already NP-hard. Impagliazzo [Imp95] described five possible worlds, from $\mathsf{P} = \mathsf{NP}$ to the existence of public-key cryptography, that are consistent with present knowledge.

2.4. **Circuits.** A Boolean circuit with n inputs is a directed acyclic graph whose sources are the inputs and whose other vertices, the gates, compute Boolean functions of their predecessors. The *size* of a circuit over the binary basis is its number of gates; for gates of larger fan-in the right measure is the number of *wires*, the sum of the fan-ins. A language $\mathcal{L}$ has circuits of size $s(n)$ if for each n some circuit of that size decides $\mathcal{L}$ on inputs of length n; the class $\mathsf{P/poly}$ consists of the languages with circuits of polynomial size. Since $\mathsf{P} \subseteq \mathsf{P/poly}$, a proof that some language in NP lies outside $\mathsf{P/poly}$ would prove $\mathsf{P} \neq \mathsf{NP}$. The circuit form is attractive because it removes uniformity, so that only combinatorics remains, and because $\mathsf{NP} \not\subseteq \mathsf{P/poly}$ is believed to be true for satisfiability itself. Kannan [Kan82] proved that $\Sigma_2\mathsf{P}$ is not contained in $\mathrm{SIZE}(n^k)$ for any fixed k, and Santhanam [San09] proved the same for Merlin–Arthur languages with one bit of advice.

2.5. **The Millennium Problem.** The Clay Mathematics Institute listed the question among its seven Millennium Prize Problems in 2000 [Clay]; the official problem description is by Cook. Fortnow [For09] and Aaronson [Aar16] survey the evidence, Wigderson [Wig19] places it within mathematics as a whole, and the textbooks of Arora and Barak [AB09] and of Jukna [Juk12] give the technical background used below. Gasarch's polls [Gas19] record a steadily growing majority of researchers who expect $\mathsf{P} \neq \mathsf{NP}$.

3. Three barriers

Three theorems show that the methods that settle other questions cannot settle this one by themselves. Each barrier is a statement about proofs, and each has been circumvented in some restricted setting.

3.1. **Relativization.** Baker, Gill and Solovay [BGS75] constructed oracles A and B with $\mathsf{P}^A = \mathsf{NP}^A$ and $\mathsf{P}^B \neq \mathsf{NP}^B$. Any argument that treats machines as black boxes, such as diagonalisation and simulation, gives the same answer relative to every oracle and therefore cannot decide the question. The hierarchy theorems relativize; the Cook–Levin theorem, which uses the local structure of computations, does not relativize in its strong forms, and the arguments of Section 5 use it through local reductions.

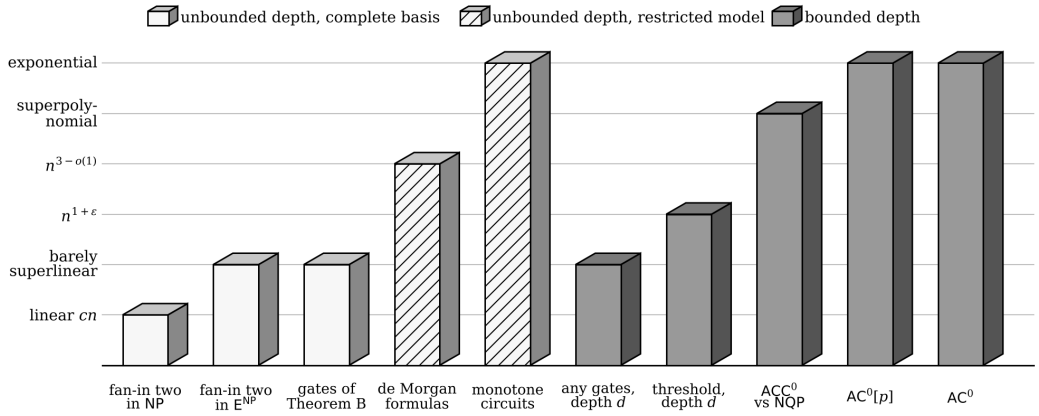


FIGURE 1. Explicit circuit lower bounds by model. The height of a bar is the order of growth of the best bound, on a scale from linear (cn) to exponential (2^{n^ϵ}); "barely superlinear" means $n \cdot \omega(1)$ but $n^{1+o(1)}$. For circuits of unbounded depth over a complete basis (orange), only linear bounds were known for explicit languages before [OAI26]; [OAI26, Corollary 1.4] and Theorem B give $cn\sqrt{\log n}$. The bounds for arbitrary gates of bounded depth are for operators with n outputs.

3.2. **Natural proofs.** Razborov and Rudich [RR97] observed that the known circuit lower bounds proceed by a property Γ of truth tables that is *useful* (no function with the property has small circuits), *large* (a fraction $2^{-O(n)}$ of all functions has it) and *constructive* (it can be decided in time polynomial in 2^n from the truth table). They proved that a natural property useful against $\mathsf{P/poly}$ would break pseudorandom functions that are secure against circuits of size 2^{n^ϵ}, whose existence follows from standard cryptographic assumptions. Arguments by diagonalisation are not natural, since they single out one function; Williams [Wil16] showed that $\mathsf{NEXP} \not\subseteq \mathcal{C}$ is in fact equivalent to the existence of a property useful against $\mathcal{C}$ that is constructive in a weak, nondeterministic sense, which locates the barrier exactly.

3.3. **Algebrization.** Aaronson and Wigderson [AW09] extended relativization to algebraic oracles, which also capture the arithmetisation used in interactive proofs. They proved that every known non-relativizing technique of that kind algebrizes, and that resolving P versus NP needs non-algebrizing methods.

3.4. **Locality.** A fourth, newer observation concerns hardness magnification. Oliveira and Santhanam [OS18], and earlier Allender and Koucký [AK10], found problems for which a slightly superlinear lower bound would imply a superpolynomial one. Chen, Hirahara, Oliveira, Pich, Rajgopal and Santhanam [CHOPRS22] proved that the known slightly superlinear lower bounds hold also for circuits with small oracle gates of unbounded function, while the magnified statements fail for such circuits; so the methods that prove the weak bounds cannot prove the magnified ones. This is the *locality barrier*.

Remark 3.1. The lower bound of Theorem B is proved by diagonalisation against a nondeterministic class combined with a satisfiability algorithm. It names a single language, so it is not natural, and it uses the local structure of computations through the reduction of Lemma 11.2. Section 12 discusses its relation to the locality barrier.

4. CIRCUIT LOWER BOUNDS KNOWN TODAY

Figure 1 arranges the results of this section by model and strength.

4.1. **General circuits.** Shannon's counting [Sha49] shows that almost every function needs about $2^n/n$ gates, and Lupanov's construction (see [Juk12, Chapter 1]) matches this. For explicit functions the only method for unbounded depth has been gate elimination: fixing an input variable removes a few gates, and repeating the step n times gives a linear bound. Schnorr [Sch80] and Blum [Blu84] pushed the constant to three (Schnorr announced $3n$, and Blum, noting a gap in one lemma, proved $3n - o(n)$ for a modification of Paul's function), Find, Golovnev, Hirsch and Kulikov [FGHK23] to $3 + \frac{1}{86}$, and Li and Yang [LY22] to $3.1n - o(n)$ for a function in P.[5] Golovnev, Hirsch, Knop and Kulikov [GHKK18] showed that natural forms of gate elimination cannot prove much more than linear bounds, and Golovnev, Kulikov, Smal and Tamaki [GKST18] turned gate elimination into satisfiability algorithms for linear-size circuits.

4.2. **Formulas.** A formula is a circuit whose gates have fan-out one. Random restrictions shrink de Morgan formulas: Håstad [Has98] proved shrinkage exponent two, so Andreev's function requires formulas of size $n^{3-o(1)}$, and Tal [Tal14] sharpened the lower-order terms to a polylogarithmic factor. Karchmer and Wigderson [KW90] identified formula depth with the communication complexity of a game; with it they proved that monotone formulas for connectivity need depth $\Omega(\log^2 n)$, and Raz and McKenzie [RM99] separated the levels of the monotone NC hierarchy.

4.3. **Bounded depth.** Ajtai [Ajt83] and Furst, Saxe and Sipser [FSS84] proved that parity is not in AC^0, the class of polynomial-size circuits of constant depth with unbounded fan-in AND and OR gates. Yao [Yao85] and Håstad [Has86] made the bound exponential: depth d requires size $2^{\Omega(n^{1/(d-1)})}$. Rossman [Ros08] proved that detecting a k-clique needs size $\omega(n^{k/4})$ at every constant depth. With parity gates added, Razborov [Raz87] and Smolensky [Smo87] proved exponential lower bounds for majority, and for counting modulo q in circuits with gates modulo a prime $p \nmid q$.

4.4. **Monotone circuits.** Razborov's method of approximations gave superpolynomial lower bounds for clique in monotone circuits, made exponential by Alon and Boppana [AB87] (see [Juk12, Chapter 9]). Tardos [Tar88] showed that the gap between monotone and general circuits can be exponential, so monotone bounds do not transfer.

4.5. ACC^0 **and the algorithmic method.** For circuits of constant depth with gates counting modulo arbitrary m, the class ACC^0, no lower bound followed from the methods above. Williams [Wil14] proved $\mathsf{NEXP} \not\subseteq \mathsf{ACC}^0$ by a satisfiability algorithm for ACC^0 circuits that beats exhaustive search, combined with the argument of [Wil13] described in Section 5. Murray and Williams [MW20] extended it to nondeterministic quasi-polynomial time, and Williams [Wil18] to ACC^0 circuits with a layer of threshold gates at the bottom.

4.6. **High classes.** Exponential classes with alternation reach maximum circuit size. Chen, Hirahara and Ren [CHR24] proved that a language in symmetric exponential time $\mathsf{S_2E}$ with one bit of advice requires circuits of size about $2^n/n$, and Li [Li24] removed the advice (the two papers appeared together in [CHLR26]); this class lies between E^{NP} and $\mathsf{ZPE}^{\mathsf{NP}}$ by Cai's theorem [Cai07]. For E^{NP} itself, whether every language has circuits of linear size was open until [OAI26, Corollary C.1].

4.7. **Threshold circuits.** A threshold gate outputs $[\sum_i w_i y_i \geq \theta]$. Polynomial-size threshold circuits of constant depth form TC^0, a class that contains all of the arithmetic of integers and for which no superpolynomial lower bound is known even at depth two for functions in NEXP. The known results concern size near linear. Hajnal, Maass, Pudlák and Szegedy [HMPST93] separated depth two from depth three for majority circuits. Impagliazzo, Paturi

[5] Over the de Morgan basis, where negations are free and the gates are AND and OR, Iwama and Morizumi [IM02] proved $5n - o(n)$.

and Saks [IPS97] proved that parity needs $n^{1+c\theta^{-d}}$ wires in threshold circuits of depth d, for constants $c > 0$ and $\theta > 1$. Kane and Williams [KW16] proved superlinear gate and superquadratic wire bounds for depth two. Chen, Santhanam and Srinivasan [CSS18] proved average-case lower bounds and satisfiability algorithms for threshold circuits of depth d with $n^{1+\epsilon_d}$ wires, and Impagliazzo, Paturi and Schneider [IPS13] gave a satisfiability algorithm for sparse depth-two threshold circuits. Chen, Tal and Wang [CTW26] proved, by Williams' method, superquadratic lower bounds for depth-two linear threshold circuits computing a function in E^{NP}. Every one of these bounds degrades with the depth and gives nothing at unbounded depth.

4.8. **Arbitrary gates.** When gates may compute any function, a single gate of fan-in n computes every Boolean function, so lower bounds concern either many outputs or bounded fan-in. For operators with n outputs, Valiant's graph-theoretic method [Val77], Pudlák's analysis of communication in bounded depth [Pud94], the bounds of Radhakrishnan and Ta-Shma for depth-two superconcentrators [RTS00], and the explicit operators of Cherukhin [Che11] and Jukna [Juk10] give superlinear wire bounds that hold only at bounded depth. Theorem B is a bound of the second kind: its table gates have fan-in at most $N^{1-\delta}$ and its depth is unrestricted.

4.9. **Algebraic circuits.** Valiant [Val79a] introduced the algebraic classes VP and VNP and showed that the permanent is complete for VNP; the conjecture $\mathsf{VP} \neq \mathsf{VNP}$ is the algebraic analogue of $\mathsf{P} \neq \mathsf{NP}$. Limaye, Srinivasan and Tavenas [LST25] proved the first superpolynomial lower bounds against algebraic circuits of every constant depth. Kabanets and Impagliazzo [KI04] showed that derandomising polynomial identity testing implies lower bounds of this kind, which ties the algebraic question to pseudorandomness.

4.10. **Range avoidance.** Given a circuit $C : \{0,1\}^n \to \{0,1\}^{n+1}$, find a string outside its range. Korten [Kor21] showed that this search problem captures explicit constructions, among them hard truth tables; Ren, Santhanam and Wang [RSW22] studied its complexity, and Ilango, Li and Williams [ILW23] connected it to indistinguishability obfuscation. The lower bounds of [CHR24, Li24] come from an algorithm for range avoidance with an NP oracle.

5. Algorithms, the exponential time hypothesis and its refutation

5.1. **Exponential algorithms for satisfiability.** Exhaustive search decides satisfiability of a formula with n variables in time 2^n times a polynomial. For k-CNF formulas much better is known. Schöning's random walk [Sch02] runs in time $(2 - 2/k)^n$; the algorithm of Paturi, Pudlák, Saks and Zane [PPSZ05] improves this, and Hertli [Her14] proved that its bounds for formulas with a unique solution hold in general, about 1.308^n for 3-SAT. Williams [Wil05] solved maximum two-constraint satisfaction in time $2^{\omega n/3}$ by fast matrix multiplication, where $\omega < 2.38$ is the exponent of matrix multiplication.

5.2. **The hypotheses.** Impagliazzo and Paturi [IP01] formulated the *exponential time hypothesis* (ETH): there is $\delta > 0$ such that 3-SAT cannot be decided in time $2^{\delta n}$. The *strong* hypothesis (SETH) asserts that for every $\epsilon > 0$ some k makes k-SAT require time $2^{(1-\epsilon)n}$. Impagliazzo, Paturi and Zane [IPZ01] proved the sparsification lemma, by which ETH is equivalent to the same statement for formulas with $O(n)$ clauses, and Calabro, Impagliazzo and Paturi [CIP06, CIP09] related clause width to clause density and extended the hypotheses to circuits of small depth. A large part of fine-grained complexity derives conditional lower bounds from these hypotheses.

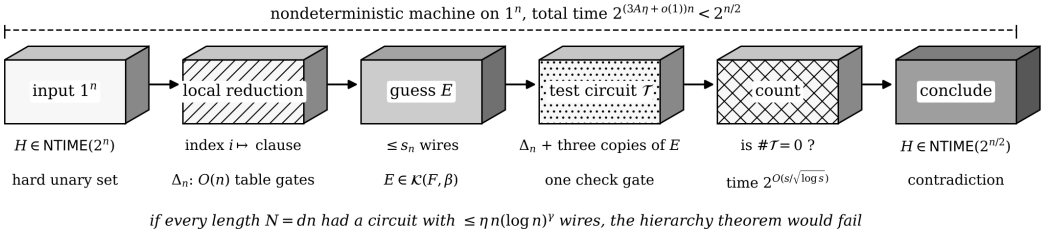


FIGURE 2. Williams' argument as used in Section 11. A unary language that is hard for nondeterministic time $2^{n/2}$ is reduced locally to a succinct 3-CNF; a small circuit for the witness language W is guessed, three copies of it read the variables of an indexed clause, and the test circuit is checked by the counting algorithm of Theorem A.

5.3. **Williams' algorithmic method.** Williams [Wil13] proved that a satisfiability algorithm for a circuit class that is only slightly faster than exhaustive search yields a lower bound for that class. The argument runs as follows (Figure 2). Suppose that every language in NEXP has small circuits in the class. By the easy-witness lemma of Impagliazzo, Kabanets and Wigderson [IKW02], every satisfiable instance then has a witness whose bits are computed by a small circuit. A nondeterministic machine guesses such a circuit, composes it with a succinct description of the instance, and verifies the witness by running the fast satisfiability algorithm on the composed circuit. This solves every language of $\mathrm{NTIME}(2^n)$ in nondeterministic time below 2^n, contradicting the nondeterministic hierarchy theorem. The survey [Wil14c] describes the many variants. The succinct description is a *local reduction*: Jahanjou, Miles and Viola [JMV18] showed that the clauses of the Cook–Levin formula for a computation of length T can be indexed by $\log T + O(\log\log T)$ bits in such a way that each clause is computed from its index by a circuit of quasi-linear size whose output bits each depend on few index bits.

Santhanam [San10] gave satisfiability algorithms for formulas, Seto and Tamaki [ST13] for formulas over the full binary basis, and Chen, Kabanets, Kolokolova, Shaltiel and Zuckerman [CKKSZ15] showed how lower-bound proofs by restriction yield such algorithms in general.

5.4. **The refutation of ETH.** In October 2026 OpenAI released the preprint [OAI26],[6] whose main theorem counts the satisfying assignments of a k-CNF with n variables and m clauses in time

$$(n+m)^{O_k(1)}\, 2^{O_k(n \log\log n/\sqrt{\log n})},$$

and in time $2^{O(n/\sqrt{\log n})}$ when $m = O(n)$. This contradicts ETH, and with the sparsification lemma it removes the basis of many conditional lower bounds. The fixed-density bound rests on a theorem about tensor networks, [OAI26, Theorem D.1], restated below as Theorem 8.3: a bipartite Boolean tensor network with $O(s)$ edges and degrees at most $s^{9/10}$ can be contracted in time $2^{O(s/\sqrt{\log s})}$. Its proof deforms the identification of the wires by a Cauchy matrix, expands the interactions between groups of vertices in variables whose two-adic valuations grow with their weight, and recovers the value at the undeformed point by Hermite interpolation with tapered multiplicities; Appendix A gives the full argument. Contraction of tensor networks had been used for the simulation of quantum circuits by Markov and Shi [MS08], with running time exponential in the treewidth; the theorem of OpenAI needs no bound on the treewidth.

Combining the fixed-density algorithm with Williams' method, [OAI26, Corollary 1.4] proves that some language in E^{NP} needs more than $cn\sqrt{\log n}$ gates of fan-in two for infinitely many n. This was the first superlinear lower bound of any kind for E^{NP}. Weighted counting,

[6] The preprint has not been refereed. Theorem A uses only its contraction theorem, whose proof is given in full in Appendix A and checked there step by step.

Max-Cut profiles and exact optimisation follow from the same algorithm [BhaHex26], and Wu and Wang [WW26] obtained bounds for extended bases of fan-in-two gates with unbounded XOR and symmetric gates (Section 11).

5.5. **What the refutation does not give.** A subexponential algorithm for 3-SAT does not place NP in P: the time $2^{O(n/\sqrt{\log n})}$ is still superpolynomial. Nor does it give a superpolynomial circuit lower bound, since Williams' method needs an algorithm whose saving over 2^n is measured against the number n of inputs of circuits of polynomial size, and the algorithms of [OAI26] are fast only when the circuit has $O(n\sqrt{\log n})$ wires.

6. Other programmes

6.1. **Geometric complexity theory.** Mulmuley and Sohoni [MS01] proposed to separate the permanent from the determinant, and so VNP from VP, by representation theory: the orbit closure of the padded permanent would not be contained in that of the determinant, and an *obstruction*, an irreducible representation occurring in the coordinate ring of one closure and not of the other, would witness this. Mulmuley [Mul12] describes the programme. Bürgisser, Ikenmeyer and Panova [BIP19] proved that occurrence obstructions, those detected by the mere presence of a representation, do not exist in the required range; obstructions by multiplicities remain possible.

6.2. **Proof complexity.** Cook and Reckhow [CR79] showed that NP $\neq$ coNP if and only if no propositional proof system has polynomial-size proofs of all tautologies; since P = NP implies NP = coNP, superpolynomial lower bounds for every proof system would separate P from NP. Haken [Hak85] proved that resolution needs exponential size for the pigeonhole principle, and Ben-Sasson and Wigderson [BSW01] related resolution size to width. Krajíček's monograph [Kra19] surveys the field; for Frege systems no superpolynomial bound is known.

6.3. **Meta-complexity.** The minimum circuit size problem MCSP asks, given a truth table and a number s, whether the function has a circuit of size s. Kabanets and Cai [KC00] began its modern study; it is in NP, and its NP-hardness is open. Hirahara [Hir23] gave a non-black-box reduction from worst case to average case for approximating it, and [Hir22] proved NP-hardness of a partial version. Ilango [Ila26] reduced SAT to MCSP relative to a random oracle, Huang, Ilango and Ren [HIR25] proved NP-hardness of approximating meta-complexity by a cryptographic approach, and Hirahara and Ilango [HI25] derived NP-hardness of MCSP from well-studied assumptions. These questions concern the complexity of complexity itself and are closely tied to natural proofs and to magnification.

6.4. **Hardness versus randomness.** Nisan and Wigderson [NW94] built pseudorandom generators from hard functions, and Impagliazzo and Wigderson [IW97] proved P = BPP if some language in E requires circuits of size $2^{\Omega(n)}$. Conversely, derandomization implies lower bounds [KI04, Wil16]. A lower bound of the strength of Theorem B is far too weak for derandomization.

6.5. **Magnification.** The results of [AK10, OS18, CHOPRS22] identify problems, many of them variants of MCSP, for which a lower bound of the form $n^{1+\epsilon}$ in some model implies a superpolynomial bound in a stronger model. The locality barrier of Section 3 explains why the known methods have not reached the magnified thresholds.

7. Repositories and the literature catalogue

7.1. **Collections of claims.** Woeginger's page [Woe16], maintained until 2016, lists the announced solutions that came to his attention, with entries dated from 1986 to 2016. Parsing the page gives 114 entries with a stated verdict: 62 claim P = NP, 49 claim P $\neq$ NP, two claim that the question is unprovable and one that it is undecidable (Figure 3). None has been accepted. The first entry already illustrates a lesson that became a theorem: polynomial

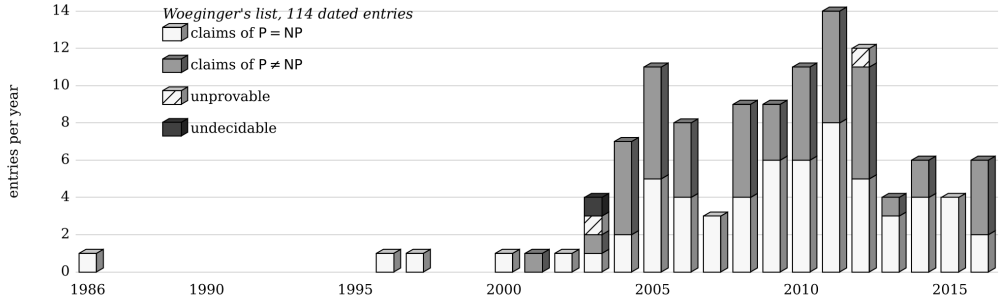


FIGURE 3. Announced solutions of the P versus NP problem listed by Woeginger [Woe16], by year and verdict (114 dated entries). None of them is accepted.

linear programs for the travelling salesman problem were refuted by Yannakakis [Yan91], who proved that every symmetric linear program expressing the problem has exponential size.

7.2. **Archives.** The Electronic Colloquium on Computational Complexity [ECCC] and the Complexity Zoo [Zoo], a catalogue of several hundred complexity classes and their known inclusions, are the standard reference points. The preprint [OAI26] and its Lean formalisation are in the public repository of OpenAI. The author's related work treats circuit architectures embedded in grids [Bha26n], quantum algorithms on symplectic state manifolds [Bha26q] and covering problems decided by exact counting over nerve complexes [Bha26c]. The Lean and Julia programs that check the finite statements of this paper are archived with it (Declarations).

7.3. **A catalogue of arXiv papers.** To measure the field, nineteen keyword queries were run against the arXiv interface, one for each topic of Sections 2 to 6: P versus NP, circuit lower bounds, natural proofs, relativization, algebrization, geometric complexity theory, hardness magnification, MCSP, meta-complexity, proof complexity, the exponential time hypotheses, threshold circuits, satisfiability algorithms for circuits, formula lower bounds, ACC^0, VP versus VNP, range avoidance, PCPs and tensor network contraction. The union contains 6,480 papers, from 1993 to the end of October 2026.[7] Table 1 gives the number of papers in the core categories by topic and period. The exponential time hypotheses (657 papers), the P versus NP question itself (436), circuit lower bounds (407) and proof complexity (332) are the largest topics; the yearly number of papers in the core set grew from 18 in 2003 to 219 in 2025 and 350 in the first ten months of 2026.

7.4. **Where the problem stands.** No proof that $\mathsf{P} \neq \mathsf{NP}$ is known, and no known method is expected to give one without new ideas, since each of the three barriers excludes a family of techniques. The strongest explicit lower bounds for unrestricted circuits are linear for functions in NP and, since [OAI26], of order $n\sqrt{\log n}$ for a function in E^{NP}. Part II extends the latter bound to circuits of threshold, exact, modular and symmetric gates and of table gates of fan-in $N^{1-\delta}$, for every $\delta > 0$.

[7] The queries combine words in the abstract with category filters, and some are broad: the words "meta" and "complexity" also match papers in machine learning. Of the 6,480 records, 1,452 have primary category `cs.CC` and 2,617 lie in the core categories of theoretical computer science and discrete mathematics; the figure and the counts below use the core set. The catalogue, the queries and the program that produced them are in the archive of the first version [Bha26v1].

TABLE 1. Papers of the arXiv catalogue in the core categories, by topic and year of first version (2,617 papers; a paper matching several topics is counted once for each; 2026 ends in October).

Topic	1993–2010	2011–15	2016–20	2021–24	2025–26	Total
exponential time hypotheses	7	79	236	189	146	657
P versus NP	96	78	117	72	73	436
circuit lower bounds	25	75	108	91	108	407
proof complexity	31	66	76	74	85	332
meta-complexity	68	53	54	95	59	329
natural proofs	32	42	63	56	55	248
formula lower bounds	26	40	56	60	43	225
PCPs	12	23	24	27	25	111
geometric complexity theory	15	20	21	23	21	100
VP versus VNP	8	17	18	10	16	69
threshold circuits	3	7	20	14	14	58
relativization	12	8	12	8	17	57
satisfiability algorithms for circuits	1	13	19	9	7	49
algebrization	8	6	4	4	11	33
MCSP	0	1	8	9	7	25
range avoidance	0	1	0	8	10	19
ACC^0	1	4	5	2	5	17
tensor network contraction	0	0	7	2	7	16
hardness magnification	0	0	3	2	2	7

8. Circuits and tensor networks

8.1. **The circuit classes.**

Definition 8.1. A *circuit* C with m inputs consists of input variables $x_1, \dots, x_m$ and gates $g_1, \dots, g_G$. Each gate g has an ordered list of $f(g) \geq 0$ predecessors, each an input or an earlier gate, and one gate is the output. The *wires* of C are the pairs (predecessor, gate), counted with multiplicity, so their number is $\mathrm{w}(C) = \sum_g f(g)$, and $s(C) = m + \mathrm{w}(C)$. For integers $F, \beta \geq 1$ the class $\mathcal{K}(F, \beta)$ consists of the circuits in which every gate g, with inputs $y = (y_1, \dots, y_f)$, is of one of the following kinds.

(a) *Table gate*: $f \leq F$, and $g(y) = \tau(y)$ for a function $\tau : \{0,1\}^f \to \{0,1\}$ given by its truth table of 2^f bits.

(b) *Weighted gate*: integers $w_1, \dots, w_f$ with $|w_i| \leq 2^\beta$ are given, $v(y) = \sum_i w_i y_i$, and $g(y) = \varphi(v(y))$, where φ is one of

$$[v \geq \theta], \qquad [v = \theta], \qquad [v \equiv \theta \bmod \mu] \ \ (1 \leq \mu \leq 2^\beta),$$

for a given integer θ, or, when $w_1 = \cdots = w_f = 1$, an arbitrary function $\varphi : \{0, \dots, f\} \to \{0,1\}$ given by its table of $f + 1$ bits.

The *description* of C lists m, the gates in order with their kinds, predecessor lists, tables, weights, thresholds and moduli in binary, and the output gate; L denotes its length. When a class parameter is written as a real power, such as $\mathcal{K}(s^{1-\delta}, s^{1-\delta})$, its integer part is meant.

A threshold θ outside $[-f2^\beta, f2^\beta + 1]$ can be replaced by the nearest end of this interval without changing the gate, and θ in a congruence can be reduced modulo μ, so we may and do assume that all numbers in a description have $O(\beta + \log f)$ bits. A table gate of fan-in zero is a constant.

Example 8.2. Every gate of fan-in at most two is a table gate. An AND or OR of any fan-in is the exact gate $[\sum_i y_i = f]$ or the threshold $[\sum_i y_i \geq 1]$; a negation is $[-y_1 \geq 0]$; parity and MOD_μ gates are congruences; majority and every symmetric function are unit-weight gates

with a table on $\{0,\dots,f\}$. Every gate of fan-in at most F, whatever it computes, is a table gate of $\mathcal{K}(F,\beta)$: in particular a linear threshold gate $[\sum_i w_i y_i \ge \theta]$ with arbitrary real w_i,θ and fan-in at most F is one, even when its minimal integer weights exceed 2^β.

8.2. **Tensor networks.** We use the definitions of [OAI26, Section D.1]. A *Boolean tensor network* over a commutative ring R is a finite multigraph $\mathcal{G}$ in which every edge carries one bit. Each vertex v has an ordered list $\delta(v)$ of incident edges, its *ports*, and a tensor $T_v : \{0,1\}^{\delta(v)} \to R$. Parallel edges occupy distinct ports. The *contraction* of the network is

$$Z = \sum_{x\in\{0,1\}^{E(\mathcal{G})}} \prod_{v\in V(\mathcal{G})} T_v(x|_{\delta(v)}).$$

The size of the network is its number of edges, and the network is *bipartite* if $\mathcal{G}$ is. For integers $P_0,\ell \ge 1$ put $B_{P_0,\ell} = (\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^\ell)$. The *description* of a network lists its graph and, for every vertex, a description of its tensor from which each entry can be computed; L is its length.

Theorem 8.3 (OpenAI, [OAI26, Theorem D.1])**.** *Fix positive constants C,A,b. Let $s \ge 2$ be an integer and let L be the description length of a bipartite tensor network with at most Cs edges, maximum degree at most $\lceil s^{9/10}\rceil$, and a number of vertices bounded by a fixed polynomial in s, whose entries lie in $B_{P_0,\ell}$ with $1 \le P_0,\ell \le Cs$. Suppose that every entry can be computed from the description in time $(s+L)^b 2^{As^{9/10}}$ and space $(s+L)^b$. Then the contraction can be computed deterministically in time $(s+L)^{O(1)}2^{O(s/\sqrt{\log s})}$ and space $(s+L)^{O(1)}$, with constants depending only on C,A,b.*

The algorithm of OpenAI is a recursion that cuts the network into 64 groups, replaces the interactions between groups by a small number of *modes*, and recovers the contraction from Taylor data at the points of a two-adic grid; it is described in full in Appendix A. Running through that proof with the degree as a parameter shows that the exponent $9/10$ enters at one place only, the inequality (A.4) that keeps the labels of the packing balanced, and that the entry cost enters only at the leaves. This gives the following form, in which U is the number of two-adic levels used at the root.

Theorem 8.4 (Parametrised contraction)**.** *For $C \ge 1$ put $U_0(C) = \lceil (C^{1/3}(17+2\log C))^{6/5}\rceil$. Let $C,b \ge 1$, and let s,U,Δ be positive integers with*

$$\text{(G1)}\ \ U \ge U_0(C), \qquad \text{(G2)}\ \ 2^{100U}\Delta \le s, \qquad \text{(G3)}\ \ 2^{4U+1} \le \Delta.$$

Let a bipartite tensor network of description length L have at most Cs edges, maximum degree at most Δ, at most s^b vertices, and entries in $B_{P_0,\ell}$ with $1 \le P_0,\ell \le Cs$, each computable from the description in time T_{ent} and space $(s+L)^b$. Then its contraction is computed deterministically in time

$$(s+L)^c\, T_{\mathrm{ent}}\, 2^{(117C+4)\,s/\sqrt{U}}$$

and space $(s+L)^c$, where c depends only on b.

The proof is in Appendix A. Theorem 8.3 is the case $\Delta = \lceil s^{9/10}\rceil$ of the next corollary.

Corollary 8.5 (Degree $s^{1-\delta}$)**.** *Fix $\delta \in (0,1)$, put $\delta' = \min(\delta,\frac12)$, and fix $C \ge 1$ and $A,b > 0$. Let $s \ge 2$, and let a bipartite tensor network with at most Cs edges, maximum degree at most $\lceil s^{1-\delta}\rceil$, at most s^b vertices and entries in $B_{P_0,\ell}$, $1 \le P_0,\ell \le Cs$, have every entry computable in time $(s+L)^b 2^{As^{1-\delta}}$ and space $(s+L)^b$. If $\log s \ge 128\,(U_0(C)+1)/\delta'$, the contraction is computed in time*

$$(s+L)^{O(1)}\, 2^{(1872C+64+A)\,s/\sqrt{\delta'\log s}}$$

and space $(s+L)^{O(1)}$. For smaller s direct summation takes time $(s+L)^{O(1)}2^{Cs+As^{1-\delta}}$, so for all $s \ge 2$ the time is $(s+L)^{O(1)}2^{O_{\delta,C,A}(s/\sqrt{\log s})}$.

TABLE 2. Running time of Theorem 8.4 against the maximum degree of the network, with $\kappa = \log(s/\Delta)$; the saving is the factor by which the exponent is below s.

Degree Δ	κ	Time exponent
$s^{9/10}$ (Theorem 8.3)	$\frac{1}{10}\log s$	$O(s/\sqrt{\log s})$
$s^{1-\delta}$, δ fixed	$\delta \log s$	$O(s/\sqrt{\delta \log s})$
$s/2^{(\log s)^{\theta}}$, $0<\theta<1$	$(\log s)^{\theta}$	$O(s/(\log s)^{\theta/2})$
$s/(\log s)^{K}$, $K>0$	$K \log\log s$	$O(s/\sqrt{K \log\log s})$
s/K_0, K_0 constant	$\log K_0$	no saving shown

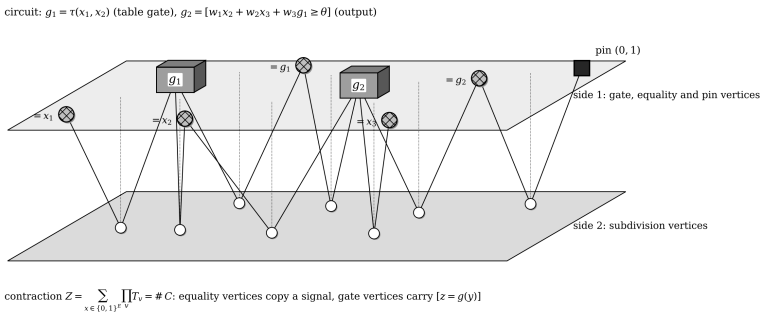


FIGURE 4. The network of a small circuit (Section 9). Gate vertices carry the tables $[z = g(y)]$; each signal is distributed to its ports by a tree of equality vertices, here a single vertex; a pin $(0, 1)$ sits on the output; every edge is then subdivided by a binary equality vertex, which makes the network bipartite.

Proof. A network of degree at most $\lceil s^{1-\delta} \rceil$ has degree at most $\lceil s^{1-\delta'} \rceil$, and its entry cost is at most $(s+L)^b 2^{As^{1-\delta'}}$, so we may assume $\delta \leq \frac{1}{2}$. Put $\lambda = \lceil \log s \rceil$, $U = \lfloor \delta\lambda/128 \rfloor$ and $\Delta = \lceil s^{1-\delta} \rceil \leq 2s^{1-\delta}$. Condition (G1) holds because $U \geq \delta \log s/128 - 1 \geq U_0(C)$. For (G2), $2^{100U} \leq 2^{100\delta(\log s+1)/128} = (2s)^{25\delta/32}$, so $2^{100U}\Delta \leq 4s^{1-7\delta/32} \leq s$ once $s^{7\delta/32} \geq 4$, which holds since $\delta \log s \geq 128$. For (G3), $4U+1 \leq \delta(\log s+1)/32+1 \leq \frac{1}{2}\log s \leq (1-\delta)\log s$. Theorem 8.4 with $T_{\mathrm{ent}} = (s+L)^b 2^{As^{1-\delta}}$ gives the exponent $(117C+4)s/\sqrt{U}$; since $\delta \log s \geq 256$ we have $U \geq \delta \log s/256$, and $16(117C+4) = 1872C + 64$. Finally $As^{1-\delta} \leq As/\sqrt{\delta \log s}$, because $s^{\delta} \geq \delta \log s$. □

When the degree is closer to s the saving falls. Write $\kappa = \log(s/\Delta)$.

Corollary 8.6 (Degree near s)**.** *Let $C \geq 1$, $\sqrt{s} \leq \Delta \leq s$ and $\kappa = \log(s/\Delta) \geq 100\,U_0(C)$. Under the other hypotheses of Theorem 8.4, the contraction is computed in time $(s+L)^{O(1)}\, T_{\mathrm{ent}}\, 2^{(1194C+41)\, s/\sqrt{\kappa}}$.*

Proof. Take $U = \lfloor \kappa/100 \rfloor$. Then (G1) holds, (G2) reads $100U \leq \kappa$, and (G3) holds because $2^{4U+1} \leq 2(s/\Delta)^{1/25} \leq 2s^{1/50} \leq \sqrt{s} \leq \Delta$ for $s \geq 5$. Since $\kappa \geq 3000$, $U \geq \kappa/104$, and $(117C+4)\sqrt{104} < 1194C + 41$. □

Table 2 lists the cases. Why the design cannot do better is explained in Section 12.2: the packing ties the number of labels to s/Δ, and the saving is the square root of the number of two-adic levels, which is at most a constant times $\log(s/\Delta)$.

9. COUNTING SATISFYING INPUTS

Theorem 9.1. *Fix $\delta \in (0, \frac{1}{2}]$. There are an absolute constant b_0 and a deterministic algorithm that, given the description of length L of a circuit $C \in \mathcal{K}(s^{1-\delta}, s^{1-\delta})$ with $s = s(C) \geq 2$,*

computes $\#C = |\{x \in \{0,1\}^m : C(x) = 1\}|$ *in time*

$$(s+L)^{b_0}\, 2^{A_0(\delta)\, s/\sqrt{\log s}}, \qquad A_0(\delta) = \frac{71500}{\sqrt{\delta}},$$

for all $s \geq 2$*, in space* $(s+L)^{b_0}$*.*

For $\delta > \frac{1}{2}$ the class is contained in the one for $\delta = \frac{1}{2}$. Since a table gate has at most $s^{1-\delta}$ inputs, $L \leq s^{O(1)} 2^{s^{1-\delta}}$, and the time is $2^{O(s/\sqrt{\delta \log s})}$. The original form of the theorem, for $\mathcal{K}(s^{4/5}, s^{2/3})$, is the case $\delta = \frac{1}{5}$. The proof occupies the rest of this section.

9.1. **Normalisation.** A gate from which the output cannot be reached may be deleted; this removes its wires and does not change the function. Constant gates of the same value may be merged, which leaves the wires unchanged. We therefore assume that every gate reaches the output and that there are at most two constant gates. Every other gate has a wire, so

$$G \leq \mathrm{w}(C) + 2. \tag{9.1}$$

Let m_0 be the number of inputs that are predecessors of no gate and that are not the output. Then $\#C = 2^{m_0}\, \#C'$, where C' is C with those inputs removed, and $s(C') \leq s(C)$. We assume $m_0 = 0$ from now on, restoring the factor at the end. A circuit is *normalised* if every gate reaches the output and there are at most two constant gates; normalising never increases the number of wires.

9.2. **The network $\mathcal{N}(C)$.** Put

$$K = \lceil s^{1-\delta/2} \rceil, \qquad \beta = \lfloor s^{1-\delta} \rfloor, \qquad \beta' = \beta + \lceil \log K \rceil + 2.$$

A gate is *small* if it is a table gate or a weighted gate of fan-in at most K, and *big* otherwise. A *signal* is an input or a gate. For a signal σ let d_σ be its number of *occurrences*: one for its own value if σ is a gate, one for each wire leaving it, and one more if it is the output.

Gate vertices. A small gate g with inputs $y_1, \dots, y_f$ becomes a vertex v_g with $f+1$ ports $(y_1, \dots, y_f, z)$ and tensor $T(y, z) = [z = g(y)]$.

Blocks. A big weighted gate g of fan-in $f > K$ has its inputs split, in order, into $k = \lceil f/K \rceil$ blocks $I_1, \dots, I_k$ of at most K inputs each. For each block there is a vertex $v_{g,j}$ with ports $(y_i)_{i \in I_j}$ and β' further ports $e_j = (e_{j,1}, \dots, e_{j,\beta'})$, and tensor

$$T(y, e_j) = \big[\, \mathrm{val}(e_j) = P_j(y) + K2^\beta \,\big], \qquad P_j(y) = \sum_{i \in I_j} w_i y_i,$$

where $\mathrm{val}(e) = \sum_t e_t 2^{t-1}$. Since $|P_j| \leq K2^\beta$, the integer $P_j + K2^\beta$ lies in $[0, 2K2^\beta]$, and $2K2^\beta < 2^{\beta'}$, so it has a unique binary expansion with β' bits. A root vertex r_g has the $k\beta'$ ports $e_1, \dots, e_k$ and one port z, with tensor

$$T(e_1, \dots, e_k, z) = \Big[\, z = \varphi\Big(\sum_{j=1}^{k} (\mathrm{val}(e_j) - K2^\beta)\Big)\Big].$$

The ports e_j of $v_{g,j}$ and of r_g are joined by β' edges, bit by bit. For a big gate, the occurrence of a wire into g is the corresponding input port of a block vertex, and the occurrence of the value of g is the port z of the root.

Equality trees and the pin. Every occurrence of a signal is a port of some vertex: an input port y_i for a wire, the port z for the value of a gate, and, for the output, the single port of a pin vertex with tensor $(T(0), T(1)) = (0, 1)$. The d_σ occurrences of σ are joined by an equality tree Q_σ: for $d_\sigma = 1$ one edge to a vertex with unary tensor $(1, 1)$; for $d_\sigma = 2$ two edges to a vertex with tensor $[a = a']$; for $d_\sigma \geq 3$ a tree with d_σ leaf edges and $d_\sigma - 2$ internal vertices of degree three, each with tensor $[a = a' = a'']$. In every case each common value on the leaf edges extends to exactly one assignment of the internal edges with all tensors equal to one, and every other assignment has product zero.

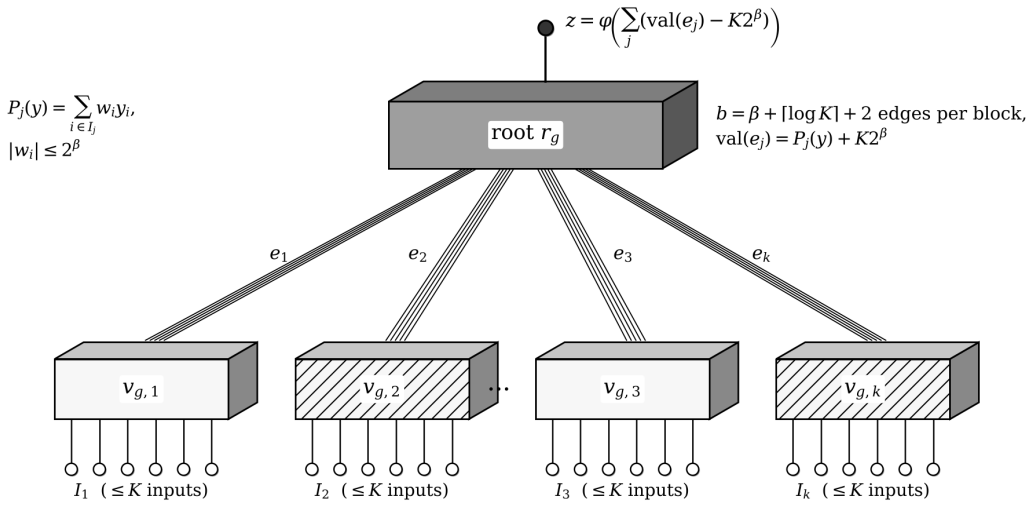


FIGURE 5. A big weighted gate. Its f inputs are cut into blocks of at most K inputs; the block vertex $v_{g,j}$ writes the shifted partial sum $P_j + K2^\beta$ on β' edges, and the root r_g adds the decoded values and applies φ. Every vertex then has degree at most $3s^{1-\delta/2}$.

Subdivision. Finally every edge is subdivided once by a vertex of degree two with tensor $[a = a']$. The vertices present before subdivision form one side and the subdivision vertices the other, so $\mathcal{N}(C)$ is bipartite. All entries are 0 or 1, read in $\mathbb{Z}/2^{m+1}\mathbb{Z}$.

Figures 4 and 5 show the construction.

Lemma 9.2 (Correctness)**.** *The contraction of $\mathcal{N}(C)$ equals $\#C$ as an integer, and hence modulo 2^{m+1}.*

Proof. Fix an assignment of the edges whose product of tensors is nonzero. The subdivision vertices force both halves of each original edge to carry the same bit. Each equality tree forces all occurrences of a signal σ to carry a common bit a_σ. At a small gate, $a_g = g(a_{\mathrm{pred}})$. At a big gate the block tensors force $\mathrm{val}(e_j) = P_j(a) + K2^\beta$, so the sum read by the root is $\sum_j P_j(a) = v(a)$ and $a_g = \varphi(v(a))$. The pin forces $a_{\mathrm{out}} = 1$. Conversely, an input x with $C(x) = 1$ determines the values of all gates, hence all bits on all edges, and the product of the tensors at that assignment is one. Distinct inputs give distinct assignments, and an input with $C(x) = 0$ gives none, because the pin would read zero. Since $m_0 = 0$, every input occurs somewhere, so the nonzero terms are in bijection with the satisfying inputs, and each is one. □

Lemma 9.3 (Size)**.** *If $\log s \ge 40/\delta$, then $\mathcal{N}(C)$ has at most $22s$ edges, at most $44s$ vertices and maximum degree at most $3s^{1-\delta/2} \le \lceil s^{1-\delta/3} \rceil$, and every entry can be computed from the description of C in time $(s + L)^{O(1)}$.*

Proof. Put $x = \log s \ge 80$. Then $x + 4 \le 2^{x/2} \le s^{1-\delta}$, so, as $\lceil \log K \rceil \le x + 1$,

$$\beta' \le s^{1-\delta} + x + 3 \le 2s^{1-\delta} \le 2^{-19} s^{1-\delta/2} \le K,$$

using $s^{\delta/2} = 2^{\delta x/2} \ge 2^{20}$.

Edges. Let $W = \mathrm{w}(C)$. The equality tree of σ has at most $2d_\sigma$ edges, and

$$\sum_\sigma d_\sigma = G + W + 1,$$

since each gate value, each wire and the output contribute one occurrence. A big gate of fan-in $f > K$ has $k \le f/K + 1 \le 2f/K$ blocks and so $k\beta' \le 2f\beta'/K \le 2f$ edges between its blocks and its root. Before subdivision the number of edges is therefore at most

$2(G+W+1)+2W \le 6W+6 \le 11s$, by (9.1) and $s \ge W$, $s \ge 2$; subdivision doubles it. Every vertex has at least one port and there are at most $2 \cdot 22s$ endpoints of edges, so there are at most $44s$ vertices.

Degrees. Subdivision vertices have degree two and equality vertices degree at most three. A table gate has degree at most $s^{1-\delta}+1$, a small weighted gate at most $K+1 \le s^{1-\delta/2}+2$, and a block vertex at most $K+\beta' \le s^{1-\delta/2}+1+2s^{1-\delta}$. For a root, $f \le s$ gives $k \le s/K+1 \le s^{\delta/2}+1$, so

$$k\beta'+1 \le (s^{\delta/2}+1)\,2s^{1-\delta}+1 = 2s^{1-\delta/2}+2s^{1-\delta}+1.$$

All of these are at most $3s^{1-\delta/2}$, and $3 \le s^{\delta/6}$ because $x \ge 6\log 3/\delta$.

Entries. An entry is a comparison of a gate value with a table entry, a weighted sum of at most s terms of $O(s)$ bits, a binary expansion or a test of φ; each takes time polynomial in $s+L$. □

Proof of Theorem 9.1. If $\log s < 66432/\delta$, enumerate the 2^m inputs; this takes time $(s+L)^{O(1)}2^s$, and $s < 258\,s/\sqrt{\delta \log s} \le A_0(\delta)s/\sqrt{\log s}$ because $\sqrt{\delta \log s} < \sqrt{66432} < 258$. Otherwise normalise C, build $\mathcal{N}(C)$ with its description of length $(s+L)^{O(1)}$, and apply Corollary 8.5 with $\delta/3$ in place of δ, $C=22$, $P_0 = m+1 \le s$, $\ell = 1$, $A = 1$ and the bounds of Lemma 9.3; the threshold of the corollary is $384(U_0(22)+1)/\delta = 66432/\delta$, since $U_0(22) = 172$. The exponent is at most

$$\frac{(1872 \cdot 22 + 65)\,s}{\sqrt{(\delta/3)\log s}} < \frac{71500\,s}{\sqrt{\delta \log s}}.$$

By Lemma 9.2 the result is $\#C' \bmod 2^{m+1}$. Multiply by 2^{m_0} modulo 2^{m+1}; since $0 \le \#C \le 2^m$, the least nonnegative residue is $\#C$. □

Remark 9.4. The algorithm uses space $(s+L)^{O(1)}$. A multitape Turing machine therefore simulates it with a factor $(s+L)^{O(1)}$ in time, by scanning its workspace at each access; this is the form used in Section 11.

Remark 9.5. The passage from δ to $\delta/3$ comes from the root of a big gate, whose degree is about $(s/K)\beta'$. A root replaced by a chain of adders, each adding one block value to a running sum of $\beta + \lceil \log s \rceil + 2$ bits, would keep the degree near 3β; since δ is arbitrary this only changes constants, and we do not use it.

Corollary 9.6. *Fix $\delta \in (0,1)$. For circuits whose gates are any mixture of arbitrary gates of fan-in at most $s^{1-\delta}$, AND, OR and parity gates of any fan-in, MOD_μ gates with μ of at most $s^{1-\delta}$ bits, symmetric gates, and threshold and exact gates with integer weights of at most $s^{1-\delta}$ bits, the number of satisfying inputs can be computed in time $L^{O(1)}2^{O_\delta(s/\sqrt{\log s})}$, where s is the number of inputs plus wires.*

Proof. These circuits lie in $\mathcal{K}(s^{1-\delta}, s^{1-\delta})$ by Example 8.2. □

10. Zero-one programs

Corollary 10.1 (Theorem C)**.** *Fix $\delta \in (0,1)$. Let $A \in \mathbb{Z}^{r\times n}$, $b \in \mathbb{Z}^r$ and $c \in \mathbb{Z}^n$, and put $\sigma = n + \nu(A)$, where $\nu(A)$ is the number of nonzero entries of A. Suppose that every entry of A and c is an integer of at most $\sigma^{1-\delta}$ bits, that is, of absolute value at most $2^{\lfloor \sigma^{1-\delta} \rfloor}$, and let L be the length of the input. Then, deterministically in time $L^{O(1)}2^{O_\delta(\sigma/\sqrt{\log \sigma})}$ and polynomial space, one can*

(i) *compute the number of $x \in \{0,1\}^n$ with $Ax \le b$;*
(ii) *decide whether such x exists and, if so, compute $\max\{c\cdot x : Ax \le b,\ x \in \{0,1\}^n\}$ and a point attaining it.*

Proof. A row with no nonzero entry is the constant condition $0 \le b_j$; if one fails, the answer is zero, and otherwise drop it. So every row has a nonzero entry and $r \le \nu(A)$. Row j becomes the weighted gate $h_j = [\sum_k (-a_{jk})x_k \ge -b_j]$, with ν_j wires, and their conjunction is

the exact gate $[\sum_j h_j = r]$ with r wires. The circuit has $s = n + \nu(A) + r \le 2\sigma$ and weights at most $2^{\lfloor \sigma^{1-\delta} \rfloor} \le 2^{\lfloor s^{1-\delta} \rfloor}$, so it lies in $\mathcal{K}(2, s^{1-\delta}) \subseteq \mathcal{K}(s^{1-\delta}, s^{1-\delta})$ and Theorem 9.1 gives (i).

For (ii) add the gate $[c \cdot x \ge t]$, with n wires, to the conjunction; the circuit now has $s \le 3\sigma$, and its weights are still at most $2^{s^{1-\delta}}$. The value $\max c \cdot x$ lies in $[-n2^{\sigma^{1-\delta}}, n2^{\sigma^{1-\delta}}]$, so binary search over t finds it with $O(\sigma^{1-\delta} + \log n)$ counting calls. A maximiser is found by fixing $x_1, x_2, \dots$ in turn, replacing b by $b - a_k x_k$ and keeping a value for which the count with $c \cdot x \ge t_{\max}$ stays positive; this uses n further calls, each on a smaller program. The calls are made one after another, so the space stays polynomial. □

When $\nu(A) = O(n)$ the bound is $2^{O_\delta(n/\sqrt{\log n})}$. Problems with $O(n)$ constraints of bounded support, such as exact cover by sets of bounded size or knapsack with conflict constraints along the edges of a sparse graph, are of this form, with weights of up to $\sigma^{1-\delta}$ bits in the knapsack row.

11. The lower bound

The argument is that of [OAI26, Appendix C.1], itself an instance of Williams' method [Wil13]; what changes is the circuit class, which is handled by Theorem 9.1 in place of the 3-CNF algorithm.

11.1. **Ingredients.** All machines are multitape Turing machines. We use three known results.

Lemma 11.1 (Nondeterministic hierarchy, [Zak83]; see [Wil13, Theorem 2.1])**.** *There is a unary language $H \subseteq \{1\}^*$ with $H \in \mathrm{NTIME}(2^n) \setminus \mathrm{NTIME}(2^{n/2})$.*

Lemma 11.2 (Local reduction, [JMV18])**.** *Let M be the machine deciding H in time 2^n. There are $q = n + O(\log n)$, $v = n + O(\log n)$, a fixed integer w, and a deterministic polynomial-time algorithm that, given n, outputs a map*

$$D_n : \{0,1\}^q \to \left(\{0,1\}^v\right)^3 \times \{0,1\}^3$$

with the following properties. Let φ_n be the 3-CNF on the variables $\{X_\lambda : \lambda \in \{0,1\}^v\}$ whose clauses are, for $i \in \{0,1\}^q$ with $D_n(i) = (\lambda_1, \lambda_2, \lambda_3, \epsilon_1, \epsilon_2, \epsilon_3)$, the clauses $X_{\lambda_1}^{\epsilon_1} \vee X_{\lambda_2}^{\epsilon_2} \vee X_{\lambda_3}^{\epsilon_3}$, where $X^1 = X$ and $X^0 = \neg X$. Then φ_n is satisfiable if and only if $1^n \in H$, and each output bit of D_n depends on at most w bits of i.

The reduction of Jahanjou, Miles and Viola [JMV18] applies to any nondeterministic computation of length T, here $T = 2^n n^{O(1)}$ including the overhead of the simulation, and gives $\log T + O(\log \log T)$ index bits and clause bits computable by decision trees of constant depth.

Lemma 11.3 (Decoder circuits)**.** *There is a circuit Δ_n that computes D_n, consists of $O(n)$ table gates of fan-in at most w, and is constructed in time polynomial in n.*

Proof. This is the argument of [OAI26, Appendix C.1]. For an output bit o of D_n, evaluate D_n at the zero string and at all strings of Hamming weight at most w; this is polynomially many evaluations since w is fixed. If o depends on coordinate c, there are two strings that differ only in c and give different values of o; setting to zero all coordinates outside the support of o keeps the values, so two such strings of weight at most w exist and are among those evaluated. Hence the evaluations recover the support of o and its truth table on it. Each output bit is then one table gate of fan-in at most w, and D_n has $O(n)$ output bits. □

11.2. **The language W.** Fix an integer $d \ge 1$ with $v \le dn$ for all large n; $d = 2$ will do. For a label $\lambda \in \{0,1\}^v$ write $\bar{\lambda} \in \{0,1\}^{dn}$ for λ followed by $dn - v$ zeros, and regard φ_n as a formula on all 2^{dn} variables $X_{\bar{\lambda}}$, most of which do not occur. For $|i| = dn$, define $\mathsf{W}(i)$ as bit i of the lexicographically first satisfying assignment of φ_n, with variables ordered by their

labels, when φ_n is satisfiable, and $\mathsf{W}(i) = 0$ otherwise; for lengths not divisible by d put $\mathsf{W}(i) = 0$.

Lemma 11.4. $\mathsf{W} \in \mathsf{E}^{\mathsf{NP}}$.

Proof. On an input of length $N = dn$, write out the 2^q clauses of φ_n by evaluating D_n, in time $2^{O(n)}$. Ask the oracle whether φ_n is satisfiable. If it is, determine the first assignment bit by bit: having fixed a prefix, ask whether φ_n with the prefix followed by zero has a satisfying extension, and append zero or one accordingly. The queries have length $2^{O(n)}$, and at most 2^{dn} of them determine bit i. The time is $2^{O(N)}$. □

11.3. **From counting to wires.** The transfer needs a little room between the class of the algorithm and the class of the lower bound, because the test circuit built from a circuit on $N = dn$ inputs is only known to have size at least n.

Theorem 11.5 (Transfer; Theorem D)**.** *Let $0 < \delta_1 < \delta_2 < 1$, let $\gamma > 0$ be rational and $A_1, b_1 > 0$, and suppose that a deterministic multitape Turing machine decides, for every circuit $C \in \mathcal{K}(s^{1-\delta_1}, s^{1-\delta_1})$ with $s = s(C) \geq 2$ given by a description of length L, whether C is satisfiable, in time $(s+L)^{b_1} 2^{A_1 s/(\log s)^\gamma}$. Let η be rational with $0 < \eta < 1/(12A_1)$. Then for infinitely many N, every circuit in $\mathcal{K}(N^{1-\delta_2}, N^{1-\delta_2})$ with N inputs deciding W on $\{0,1\}^N$ has more than $\eta N(\log N)^\gamma/(2d)$ wires.*

Proof. Put $s_n = \lfloor \eta n \lfloor \lfloor \log n \rfloor^\gamma \rfloor \rfloor$, which is computable in time polynomial in $\log n$, and $F_n = \lfloor (dn)^{1-\delta_2} \rfloor$. Suppose, for contradiction, that for all large n some circuit $E_n \in \mathcal{K}(F_n, F_n)$ with dn inputs and at most s_n wires decides W on $\{0,1\}^{dn}$; normalising E_n keeps it in the class and does not add wires. We decide H as follows. Reject inputs that are not unary. On 1^n, guess a normalised circuit $E \in \mathcal{K}(F_n, F_n)$ with dn inputs and at most s_n wires. By (9.1) it has at most $s_n + 2$ gates, each with a table of at most 2^{F_n} bits or at most s_n numbers of $O(F_n + \log n)$ bits, so its description has length $L_E = 2^{O(n^{1-\delta_2})}$. Build the test circuit $\mathcal{T}$ with inputs $i \in \{0,1\}^q$:

(1) the decoder Δ_n of Lemma 11.3 computes $(\lambda_1, \lambda_2, \lambda_3, \epsilon_1, \epsilon_2, \epsilon_3)$;
(2) three copies of E read $\bar{\lambda}_1, \bar{\lambda}_2, \bar{\lambda}_3$, the padding zeros coming from one constant gate;
(3) one table gate of fan-in six outputs one if and only if every literal $E(\bar{\lambda}_t)^{\epsilon_t}$ is false, that is, if the clause of index i is violated by the assignment $\lambda \mapsto E(\bar{\lambda})$.

Accept if and only if $\mathcal{T}$ is unsatisfiable.

If $1^n \in H$, then φ_n is satisfiable and the guess $E = E_n$ computes a satisfying assignment, so no clause is violated and the machine accepts. If $1^n \notin H$, then every E defines an assignment that violates some clause of φ_n, so $\mathcal{T}$ is satisfiable and the machine rejects. Thus the machine decides H.

The circuit $\mathcal{T}$ has $q \geq n$ inputs and at most $3s_n + O(n)$ wires, so $n \leq s(\mathcal{T}) \leq 3s_n + B_0 n$ for a constant B_0. Its table gates have fan-in at most $\max(w, 6, F_n)$ and its weights and moduli are at most 2^{F_n}. For $n \geq d^{(1-\delta_2)/(\delta_2-\delta_1)}$ we have $(dn)^{1-\delta_2} \leq n^{1-\delta_1}$, and $\max(w, 6) \leq n^{1-\delta_1}$ for large n; since $s(\mathcal{T}) \geq n$, the circuit lies in $\mathcal{K}(s^{1-\delta_1}, s^{1-\delta_1})$ with $s = s(\mathcal{T})$. By hypothesis its satisfiability is decided in time

$$(s + L_{\mathcal{T}})^{b_1} 2^{A_1 s/(\log s)^\gamma} \leq 2^{O(n^{1-\delta_2})} \cdot 2^{A_1(3s_n + B_0 n)/(\log n)^\gamma} = 2^{(3A_1\eta + o(1))n},$$

and $3A_1\eta < 1/4$. Guessing E and building $\mathcal{T}$ take $2^{O(n^{1-\delta_2})}$ steps. Hence $H \in \mathrm{NTIME}(2^{n/2})$ after hardwiring finitely many lengths, which contradicts Lemma 11.1.

Therefore, for infinitely many n, every circuit in $\mathcal{K}(F_n, F_n)$ deciding W on length $N = dn$ has more than s_n wires. Since $\log n = \log N - \log d$, we have $s_n \geq (\eta/d)(1 - o(1))N(\log N)^\gamma$, which exceeds $\eta N(\log N)^\gamma/(2d)$ for large N. □

The condition $\delta_1 < \delta_2$ is used only for the membership of $\mathcal{T}$: with $\delta_1 = \delta_2$ the inequality $(dn)^{1-\delta_2} \leq s^{1-\delta_1}$ would need $s(\mathcal{T}) \geq dn$, which fails in general. The guess costs $2^{O(N^{1-\delta_2})}$,

which is $2^{o(n)}$; this step alone would tolerate any fan-in that is $o(N)$, and the binding step is the satisfiability algorithm.

Theorem 11.6 (Theorem B)**.** *There is a language* $\mathsf{W} \in \mathsf{E}^{\mathsf{NP}}$ *with the following property. For every* $\delta \in (0,1)$ *and infinitely many* N*, every circuit in* $\mathcal{K}(N^{1-\delta}, N^{1-\delta})$ *with* N *inputs that decides* W *on* $\{0,1\}^N$ *has more than* $c(\delta)N\sqrt{\log N}$ *wires, where*

$$c(\delta) = \frac{\sqrt{\delta'}}{5 \cdot 10^6}, \qquad \delta' = \min(\delta, \tfrac{1}{2}).$$

Proof. The class shrinks as δ grows, so we may assume $\delta \le \frac{1}{2}$. Theorem 9.1 with $\delta/2$ in place of δ, together with Remark 9.4, decides satisfiability of circuits in $\mathcal{K}(s^{1-\delta/2}, s^{1-\delta/2})$, by testing whether $\#C > 0$, on a multitape machine in time $(s+L)^{b_0+O(1)}2^{A_1 s/\sqrt{\log s}}$ for large s, with $A_1 = A_0(\delta/2) = 71500\sqrt{2}/\sqrt{\delta} < 101200/\sqrt{\delta}$; finitely many sizes are absorbed in the constant b_1. Apply Theorem 11.5 with $\delta_1 = \delta/2$, $\delta_2 = \delta$, $\gamma = \frac{1}{2}$ and $d = 2$, and choose a rational η with $4c(\delta) < \eta < \sqrt{\delta}/(12 \cdot 101200)$; this is possible since $4/(5 \cdot 10^6) < 1/1214400$. The bound is $\eta N\sqrt{\log N}/4 > c(\delta)N\sqrt{\log N}$. Lemma 11.4 gives $\mathsf{W} \in \mathsf{E}^{\mathsf{NP}}$. □

The language W does not depend on δ, so one language satisfies the bound for every fixed δ, each time for an infinite set of lengths that may depend on δ. The case $\delta = \frac{1}{4}$ is the original form: table gates of fan-in $N^{3/4}$ and weights of $N^{3/4}$ bits, more than the $N^{1/2}$ bits of the first version. The chain of exponents behind Theorem 11.6 is: the lower bound at δ uses counting at $\delta/2$, which uses contraction at degree $s^{1-\delta/6}$, which uses $U = \lfloor \delta\lambda/768 \rfloor$ two-adic levels.

Corollary 11.7 (Mixed gates)**.** *Fix* $\delta \in (0,1)$*. For infinitely many* N*, the language* W *on* $\{0,1\}^N$ *is not decided by any circuit with at most* $c(\delta)N\sqrt{\log N}$ *wires whose gates are, in any combination and at any depth: arbitrary Boolean gates of fan-in at most* $N^{1-\delta}$*; AND, OR, parity and symmetric gates of any fan-in;* MOD_μ *gates with* μ *of at most* $N^{1-\delta}$ *bits; and threshold and exact gates of any fan-in with integer weights of at most* $N^{1-\delta}$ *bits.*

Proof. Such circuits lie in $\mathcal{K}(N^{1-\delta}, N^{1-\delta})$ by Example 8.2. □

Corollary 11.8 (Threshold circuits of fan-in $N^{1-\delta}$)**.** *Fix* $\delta \in (0,1)$*. For infinitely many* N*, every circuit of linear threshold gates* $[\sum_i w_i y_i \ge \theta]$ *with arbitrary real* w_i *and* θ*, of any depth, in which every gate has fan-in at most* $N^{1-\delta}$*, that decides* W *on* $\{0,1\}^N$ *has more than* $c(\delta)N\sqrt{\log N}$ *wires.*

Proof. A gate of fan-in $f \le N^{1-\delta}$ computes a function $\{0,1\}^f \to \{0,1\}$ and is therefore a table gate with the same predecessors (Example 8.2). The circuit lies in $\mathcal{K}(N^{1-\delta}, N^{1-\delta})$ with the same wires. □

No bound on the weights is used here. Muroga's theorem, that a threshold function of f variables has integer weights of $O(f \log f)$ bits [Mur71, Has94], gives a second route, through the weighted gates of $\mathcal{K}(2, N^{1-\delta/2})$, but it is not needed.

For fan-in closer to N the same argument, with Corollary 8.6 in place of Corollary 8.5, still gives a superlinear bound, with a smaller factor.

Proposition 11.9 (Fan-in $N/(\log N)^K$)**.** *For every* $K > 1$ *there is* $c > 0$ *such that for infinitely many* N *every circuit in* $\mathcal{K}(N/(\log N)^K, N/(\log N)^K)$ *with* N *inputs deciding* W *on* $\{0,1\}^N$ *has more than* $cN\sqrt{\log\log N}$ *wires.*

Proof. Put $\ell = \log s$ and $K' = K - 1$. For counting in $\mathcal{K}(s/\ell^{K'}, s/\ell^{K'})$ build $\mathcal{N}(C)$ with blocks of size $\lceil s/\ell^{K'/2} \rceil$ and $\beta = \lfloor s/\ell^{K'} \rfloor$. As in Lemma 9.3, β' is at most the block size, $\mathcal{N}(C)$ has at most $22s$ edges, and its degree is at most $\Delta = 3s/\ell^{K'/2}$, for large s. Corollary 8.6 with $\kappa = \frac{K'}{2}\log\ell - \log 3$ counts in time $(s+L)^{O(1)}2^{O(s/\sqrt{K'\log\log s})}$ once $\kappa \ge 17200$, and by enumeration below. In the transfer use the saving $g(s) = \sqrt{\log\log s}$ in place of $(\log s)^\gamma$

and $s_n = \lfloor \eta n \lfloor \sqrt{\lfloor \log \lfloor \log n \rfloor \rfloor} \rfloor \rfloor$. Membership of $\mathcal{T}$ needs $dn/(\log dn)^K \le n/(\log n)^{K'}$, which holds for $\log n \ge d$; the guess and the factor $(s+L)^{b_1}$ cost $2^{O(n/(\log n)^K)} = 2^{o(n)}$; and $2^{A_1 s/g(s)} \le 2^{A_1(3s_n+B_0 n)/g(n)} = 2^{(3A_1\eta+o(1))n}$. The rest of the proof of Theorem 11.5 is unchanged. □

11.4. **Comparison.** For fan-in-two circuits, Theorem B with any δ contains [OAI26, Corollary 1.4], since g such gates use at most $2g$ wires. Wu and Wang [WW26], in a preprint that appeared two days before the first version of this paper, prove for a language in E^{NP} a lower bound of $\Omega(m\sqrt{\log m}/\log\log m)$ *gates* for circuits of unbounded fan-in AND and OR gates with binary XOR gates, and $\Omega(m\sqrt{\log m})$ gates for the binary basis with a linear budget of unbounded XOR and symmetric gates. A bound on gates for unbounded fan-in AND and OR is not implied by a bound on wires, and conversely their bases have no table gates of large fan-in and admit only a budget of large XOR and symmetric gates; the two sets of results are incomparable. For threshold circuits, the bounds of [IPS97, KW16, CSS18, CTW26] hold for functions in P or are stronger at each fixed depth, but they tend to linear as the depth grows; Theorem B is weaker at small depth and holds at every depth. To the author's knowledge no superlinear lower bound for a language in E^{NP} against unbounded-depth circuits of arbitrary gates of fan-in $N^{1-\delta}$, or of threshold and MOD_μ gates without a budget, was known before.

12. Limits of the method

12.1. **The exponent $\sqrt{\log N}$.** Theorem 11.5 converts a saving of $(\log s)^\gamma$ in the exponent of a satisfiability algorithm into a factor $(\log N)^\gamma$ in the lower bound, and conversely the method can give no more: the test circuit has $\Theta(s_n)$ wires and must be checked in time $2^{n/2}$. In the proof of Theorem 8.4 the factor $\sqrt{U}$ has a definite source (Appendix A.5). At a call with M edges, an interpolation node of two-adic level i has at most $6M/D$ modes of each weight, $D = \Lambda/2^i$, and works at a precision of at most $34MD/u$, where u is the number of retained levels; the number of admissible mode monomials is then $\exp(O(\sqrt{M \cdot MD/u \cdot D^{-1}})) = \exp(O(M/\sqrt{u}))$, whatever D is. The recursion divides M by eight at each depth while u halves, and these exponents sum to $O(s/\sqrt{U})$. A bound $cN\log N$ would need a contraction algorithm with exponent $O(s/\log s)$, that is, either U of order $(\log s)^2$ or a count of monomials that does not lose the square root.

12.2. **The design constraint.** The number U of levels cannot exceed a constant times $\log(s/\Delta)$ in this design, for three reasons that are tied together.

(i) *Degree against labels.* A vertex of degree Δ with row label ρ sends all its edges to the r^a pair labels $(\rho,\cdot)$, so some pair-label class receives at least Δ/r^a edges whatever the labelling. Classes are padded to a common size, and there are r^{2a} of them, so the packets have at least Δr^a slots. Property (1) of Lemma A.1, at most $3M$ slots, forces $r^a \le 3M/\Delta$, that is $a \le \frac{1}{6}\log(3M/\Delta)$. The second-moment labelling achieves $a \le \frac{1}{24}\log(M/2\Delta)$; a sharper balancing could gain a constant factor in a, and no more.

(ii) *Levels against packet length.* The levels are two-adic valuations of points of S, so $u \le a$; and the inherited precision is dominated only if $D_t \ge 2^{a_{d-1}+u}$, which forces $a_d \ge a_{d-1} + 2u_d$.

(iii) *Exponent against levels.* At level i there are $\Theta(M/D)$ modes of each weight and precision $\Theta(MD/u)$, which gives $\exp(\Theta(M/\sqrt{u}))$ monomials in the count above, independently of D.

Together these give the exponent $\Theta(s/\sqrt{U})$ with $U \le a_0 = O(\log(s/\Delta))$, which is Table 2. The packet tensors, which cost $\exp(O(\Lambda\log s)) = \exp(s^{o(1)})$ to evaluate, are not what limits a; the balance of the packing is. Whether a different packing, for instance one that treats vertices of high degree separately, or a different algorithm removes the constraint is open. There is no obstruction from the problem itself: one vertex of degree s with a table of 2^s entries is contracted in time polynomial in its description.

12.3. **Fan-in and weights.** Some restriction on the table gates is unavoidable, since one gate of fan-in N decides any language. In the proof, a table gate of fan-in F costs 2^F in the guess and in the entries of the network. The guess tolerates any $F = o(N)$; the network tolerates entry costs of $2^{O(s/\sqrt{\log s})}$; the binding constraint is the degree Δ of the network, which must satisfy $\log(s/\Delta) = \Omega(\log s)$ for the full factor $\sqrt{\log N}$. This is why Theorem 11.6 holds for fan-in $N^{1-\delta}$ and Proposition 11.9 gives only $\sqrt{\log\log N}$ for fan-in $N/(\log N)^K$. Between fan-in $N/(\log N)^K$ and fan-in N/K_0 the method gives no superlinear bound. Weights enter through the number β' of bits per block, which must keep the degree of the roots below Δ.

12.4. **Tree encodings of weighted gates.** The network $\mathcal{N}(C)$ replaces a weighted gate by a star: block vertices joined to a root by bundles of β' edges. The root has degree about $f\beta'/K$, so a gate of fan-in f with weights of β bits is encoded within degree Δ only when β is at most of the order of Δ (Lemma 9.3 and Remark 9.5). The following theorem, which uses no unproved statement, shows that no encoding whose graph is a tree does essentially better: some exact gates with weights of fewer than $2f$ bits need degree $f/(4\log^3 f)$ in every tree network.

An *open network* over a field $\mathbb{F}$ is a network in the sense of Section 8 in which some edges, the *legs*, have only one end. If U is the set of legs, its tensor is $h(x_U) = \sum_x \prod_v T_v(x|_{\delta(v)})$, the sum running over the values of the other edges. It is a *tree network* if the simple graph Γ on its vertices, two vertices being adjacent when some edge joins them, is a forest. Its *degree* is the largest number of ports of a vertex, legs and parallel edges included. For $h : \{0,1\}^U \to \mathbb{F}$ and $A \subseteq U$ let $\mathrm{rk}_A(h)$ be the rank of the $2^{|A|} \times 2^{|U\setminus A|}$ matrix with entries $h(x_A, x_{U\setminus A})$.

Lemma 12.1. *Let a tree network with legs U have tensor h, let v be a vertex, and let Q be a connected component of $\Gamma - v$ that contains the legs A and is joined to v by $\mu \geq 0$ edges. Then $\mathrm{rk}_A(h) \leq 2^\mu$.*

Proof. Every edge leaving Q ends at v. Summing first over the edges inside Q and then over the others gives $h(x) = \sum_{e\in\{0,1\}^\mu} P(x_A, e)\, R(e, x_{U\setminus A})$, a sum of 2^μ matrices of rank at most one. □

Let $G = (V, E)$ be a simple graph with edges $e_1, \dots, e_k$. The gate I_G has inputs x_a for $a \in V$ and y_j for $1 \leq j \leq k$, so its fan-in is $f = |V| + |E|$, and

$$I_G(x, y) = \prod_{j\colon e_j = \{a,b\}} [x_a + x_b + y_j = 1].$$

Give x_a the weight $\sum_{j\colon a\in e_j} 4^{j-1}$ and y_j the weight 4^{j-1}, and put $\theta = \sum_j 4^{j-1}$. The $(j-1)$-st base-four digit of the weighted sum is $x_a + x_b + y_j \leq 3$ for $e_j = \{a, b\}$, so there are no carries, and I_G is the exact gate $[\sum w \cdot (x, y) = \theta]$ with positive weights below 4^k, that is, of at most $2k$ bits. Call $e_j = \{a, b\}$ *split* by a partition $A \sqcup B$ of the inputs if the three inputs x_a, x_b, y_j do not all lie on one side.

Lemma 12.2. *If M is a set of split edges forming an induced matching of G (no two share a vertex and no edge of G joins endpoints of two of them), then $\mathrm{rk}_A(I_G) \geq 2^{|M|}$ over every field.*

Proof. For $e \in M$ choose an input $p_e \in A$ and an input $q_e \in B$ among the three inputs of e, and call the third one r_e. For $\tau \in \{0,1\}^M$ define an assignment α^τ of A and γ^τ of B as follows: $p_e = 1 - \tau_e$, $q_e = \tau_e$ and $r_e = 0$ for $e \in M$; $x_c = 0$ for every vertex c not covered by M; $y_j = 1$ for every edge $e_j \notin M$ with no endpoint covered by M; and $y_j = 1 - x_a$ for an edge $e_j = \{a, c\} \notin M$ with a covered by M, where x_a is the value that α^τ or γ^τ, whichever assigns y_j, gives to x_a. Since M is induced, such an edge has c uncovered, and no edge outside M joins two covered vertices. Every constraint holds under $(\alpha^\tau, \gamma^\tau)$, so $I_G(\alpha^\tau, \gamma^\tau) = 1$. If $\tau \neq \sigma$, choose $e \in M$ with $\tau_e \neq \sigma_e$; under $(\alpha^\tau, \gamma^\sigma)$ the constraint of e reads $(1 - \tau_e) + \sigma_e \in \{0, 2\}$, so $I_G(\alpha^\tau, \gamma^\sigma) = 0$. The matrix of I_G therefore contains an identity matrix of order $2^{|M|}$. □

Lemma 12.3. *Let G have maximum degree $d \geq 1$ and edge expansion $h > 0$, that is, $e(S, V \setminus S) \geq h|S|$ whenever $|S| \leq |V|/2$. For every set A of inputs, $A \sqcup B$ has at least $c_1 \min(|A|, |B|)$ split edges, with $c_1 = \min(\frac{1}{4}, \frac{h}{d+1})$, and an induced matching of at least $c_1 \min(|A|, |B|)/(2d^2)$ split edges.*

Proof. Write A_V, A_E for the vertex and edge inputs in A, and B_V, B_E likewise. By symmetry we may assume $|A_V| \leq |V|/2$ and prove the bound $c_1|A|$; if instead $|B_V| < |V|/2$, the same argument gives $c_1|B|$. The edges of G between A_V and B_V are split, and there are at least $h|A_V|$ of them. An edge input $y_j \in A_E$ whose edge has an endpoint in B_V is split; at most $d|A_V|/2$ edges have both endpoints in A_V, so there are at least $|A_E| - d|A_V|/2$ of these. If $|A_E| \geq d|A_V|$, this is at least $|A_E|/2 \geq |A|/4$. Otherwise $|A| < (d+1)|A_V|$, and the first count is at least $h|A|/(d+1)$. For the matching choose split edges greedily: a chosen edge $\{a, b\}$ excludes only the edges meeting the at most $2d$ vertices adjacent to a or b or equal to them, at most $2d^2$ edges. □

Theorem 12.4. *Let G be a simple graph with maximum degree d and edge expansion $h > 0$, and $f = |V| + |E|$. Over every field, every tree network whose tensor is I_G has degree at least $c_G f$, where $c_G = \min(\frac{1}{4}, \frac{h}{d+1})/(2d^2)$. For the k-dimensional cube $G = Q_k$ the gate I_G has fan-in $f = 2^k + k2^{k-1}$ and weights of fewer than $2f$ bits, and every tree network for it has degree at least $f/(4 \log^3 f)$.*

Proof. Let $\ell(v)$ be the number of legs at v. Some vertex v has the property that every component of $\Gamma - v$ contains at most $f/2$ legs: if no tree of Γ has more than $f/2$ legs any vertex will do, and otherwise one walks in that tree towards the unique component with more than $f/2$ legs, which never returns and so stops. Let $Q_1, \ldots, Q_t$ be the components of $\Gamma - v$, with leg sets A_i, $|A_i| \leq f/2 \leq f - |A_i|$, joined to v by μ_i edges. By Lemmas 12.1, 12.2 and 12.3, $2^{\mu_i} \geq \mathrm{rk}_{A_i}(I_G) \geq 2^{c_G|A_i|}$. Summing over i,

$$c_G(f - \ell(v)) \leq \sum_i \mu_i \leq \deg v - \ell(v),$$

so $\deg v \geq c_G f + (1 - c_G)\ell(v) \geq c_G f$. For the cube, $d = k$ and the edge-isoperimetric inequality $e(S) \leq \frac{1}{2}|S| \log |S|$ for $S \subseteq \{0,1\}^k$ gives $e(S, \bar{S}) = k|S| - 2e(S) \geq |S|(k - \log |S|) \geq |S|$ when $|S| \leq 2^{k-1}$, so $h \geq 1$. (The inequality follows by induction on k: splitting S along a coordinate into parts of sizes $a \leq b$ gives $e(S) \leq e(S_0) + e(S_1) + a$, and $a \log a + b \log b + 2a \leq (a+b) \log(a+b)$, with equality at $a = b$ and a difference that does not decrease in b.) Hence $c_G = 1/(2k^2(k+1)) \geq 1/(4k^3)$, $k \leq \log f$, and the weights have $2|E| = k2^k < 2f$ bits. □

With the degree bound Δ of Corollary 8.5, a star of blocks encodes every weighted gate whose weights have at most a constant times Δ bits. By Theorem 12.4 no tree encoding of degree Δ handles every exact gate with weights of $O(\Delta \log^3 \Delta)$ bits: take the least k with $f_k > 4\Delta \log^3 f_k$, where $f_k = 2^{k-1}(k+2)$ is the fan-in of I_{Q_k}; then $f_{k-1} \leq 4\Delta \log^3 f_{k-1}$, so $f_{k-1} = O(\Delta \log^3 \Delta)$, and the weights have fewer than $2f_k \leq 6f_{k-1}$ bits. The weight restriction of Theorem A is thus necessary for encodings of this shape, up to the logarithmic factor. The theorem says nothing about networks whose graphs are not trees: I_G itself, a conjunction of $|E|$ constraints on three bits each, has a network with $O(f)$ edges and degree $d + 1$, and cut-rank arguments cannot settle the question, since an expander network has every balanced cut of order f with only $O(f)$ edges. A general threshold gate has an integer representation with weights of $O(f \log f)$ bits [Mur71], and for $f\beta \gg \Delta$ neither an encoding with $O(f)$ edges and degree Δ nor a proof that none exists is known.

12.5. **Superpolynomial bounds.** Williams' theorem [Wil13] converts an algorithm for circuits of polynomial size with n inputs into a lower bound only if it beats 2^n by a superpolynomial factor. Theorem 9.1 is measured in the size: for a circuit with n inputs and n^k wires it gives time $2^{O(n^k/\sqrt{\log n})}$, which is far above 2^n. Superpolynomial lower bounds for

E^{NP}, and a fortiori for NP, need algorithms whose running time depends on the number of inputs; nothing in the tensor method provides this.

12.6. **From E^{NP} to NP.** The language W is defined by the first satisfying assignment of an exponentially long formula and is computed with an NP oracle in exponential time; it is not known to lie in NP. The lower bound for W therefore says nothing about P versus NP directly, and even a superpolynomial lower bound for E^{NP} would leave the question open, since $\mathsf{P} \neq \mathsf{NP}$ is not implied by lower bounds for exponential-time classes.

12.7. **The barriers.** The proof diagonalises against $\mathrm{NTIME}(2^{n/2})$ through a single language, so it is not a natural proof in the sense of [RR97]. Because the table gates may compute arbitrary functions of $N^{1-\delta}$ inputs, Theorem B holds for circuits with gates of arbitrary function and almost linear fan-in. The locality barrier [CHOPRS22] shows that the known proofs of lower bounds extend to circuits with small oracle gates, while the magnified statements of [OS18] are false for such circuits; an argument that applies equally to gates of arbitrary function, as this one does, is not expected to reach them. The barrier concerns techniques and does not formally exclude any extension, but it indicates that the route from Theorem B to superpolynomial bounds does not pass through magnification.

Appendix A. Proof of the contraction theorem

This appendix proves Theorem 8.4, and with it Theorem 8.3 and Corollaries 8.5 and 8.6. The algorithm is that of OpenAI [OAI26, Appendix D.4]; the presentation follows theirs step by step, with the number U of two-adic levels and the degree Δ kept as parameters and every constant written out. Throughout, log is to base two and ln is natural. A call of the recursion receives a bipartite network with M edges and a working precision p, and returns its contraction modulo 2^p. One side of the network is called the row side and the other the column side; the ports of row vertices are *row ports* and those of column vertices *column ports*. To keep the two lengths apart, L is the description length of the input network and Λ the length of a packet.

A.1. **The coefficient ring.** Put $\lambda = \lceil \log s \rceil$ and $e = 4\lambda$. Choose a monic $f \in \mathbb{Z}[\xi]$ of degree e with coefficients in $\{0,1\}$ whose reduction modulo two is irreducible; such f exists because $\mathbb{F}_{2^e}$ exists, and it is found in time polynomial in s by testing the $2^e = s^{O(1)}$ candidates for divisibility by every monic polynomial of degree at most $e/2$ over $\mathbb{F}_2$ [vzGG13]. Put

$$\mathcal{O} = \mathbb{Z}_{(2)}[\xi]/(f), \qquad \mathbb{K} = \mathbb{Q}[\xi]/(f), \qquad \mathcal{A} = \mathcal{O}[V]/(V^{\ell}), \qquad \mathcal{A}_p = \mathcal{A}/(2^p),$$

where $\mathbb{Z}_{(2)}$ is the ring of rationals with odd denominator. Since f is irreducible modulo two it is irreducible over $\mathbb{Q}$, so $\mathbb{K}$ is a field. For $x \in \mathbb{K}^{\times}$ let $v_2(x)$ be the least two-adic valuation of its coordinates in the basis $1, \xi, \dots, \xi^{e-1}$. Then $\mathcal{O}/2\mathcal{O} = \mathbb{F}_{2^e}$, and factoring out the least coordinate valuation shows $v_2(xy) = v_2(x) + v_2(y)$. An element of $\mathcal{O}$ that is nonzero modulo two is a unit, since the determinant of its multiplication matrix is odd. Arithmetic in $\mathcal{A}_p$, with coordinates in the bases $1, \xi, \dots, \xi^{e-1}$ and $1, V, \dots, V^{\ell-1}$ stored as residues in $[0, 2^p)$, takes time and space polynomial in s, p, ℓ; units are inverted by inversion modulo two and Newton lifting. Divisibility in $\mathcal{A}$ is coordinatewise.

The lifts to $\mathcal{O}$ of the $2^e \geq s^4$ elements of $\mathbb{F}_{2^e}$, taken with coordinates in $\{0,1\}$, have pairwise differences that are units; Lagrange interpolation at fewer than s^4 of them is therefore valid over every $\mathcal{A}_p$. The original entries lie in $B_{P_0,\ell} \subseteq \mathcal{A}_{P_0}$, and the contraction, a polynomial in the entries, is computed in $\mathcal{A}_{P_0}$ and read off in $B_{P_0,\ell}$.

When the precision grows, each tensor entry is lifted separately: an entry specified modulo 2^{p_c} is replaced by the element of $\mathcal{A}$ whose coordinates are its integer representatives in $[0, 2^{p_c})$. This lift is not additive, but it is fixed: every tensor carries the precision p_c at which it was created, and every later evaluation uses the same representatives. Precisions strictly increase along a recursive path, by Lemma A.3 below, so the representatives are canonical at every later precision.

A.2. **Parameters and packing.** Let C, b, s, U, Δ satisfy (G1)–(G3) of Theorem 8.4. Put

$$H = \Big\lceil \frac{s}{\lfloor \sqrt{U} \rfloor} \Big\rceil, \qquad r = 64.$$

A call with $M \le H$ edges is a *leaf* and is evaluated by summing over all 2^M edge assignments; a call with $M > H$ is *active*. At an active call at depth d (the root has depth zero) put

$$u = u_d = \Big\lfloor \frac{U}{2^d} \Big\rfloor, \qquad a = a_d = 2\sum_{j=0}^{d} u_j, \qquad \Lambda = 2^a, \qquad D_{\mathrm{arr}} = r^{2a} = 2^{12a}, \tag{A.1}$$

and $a_{-1} = 0$. Coordinates of labels and bits of slot numbers are numbered $0, \dots, a-1$, bit c having weight 2^c. It will be shown in Section A.6 that every child of a call with M edges has at most $M/8$ edges and maximum degree at most Δ. We record the consequences.

Size of s. By (G2), $\log s \ge 100U$.

Depth. On an active path $s/\lfloor \sqrt{U} \rfloor \le H < M \le Cs8^{-d}$, so

$$8^d < C\sqrt{U}, \qquad 2^d < C^{1/3}U^{1/6}. \tag{A.2}$$

Levels. By (A.2) and (G1), $U/2^d > U^{5/6}C^{-1/3} \ge 17 + 2\log C$. Hence on every active call

$$u \ge 16 + 2\log C \ge 16, \qquad u \ge \frac{16}{17}\cdot\frac{U}{2^d}, \qquad \sqrt{U} < C^{1/5}(u+1)^{3/5}, \tag{A.3}$$

the last because $U < (u+1)2^d < (u+1)C^{1/3}U^{1/6}$. Moreover $2u_d \le a_d \le 4U$, so $u \le a$, $a \ge 32$, and

$$\Lambda \le 2^{4U} \le s^{1/25}, \qquad D_{\mathrm{arr}} \le 2^{48U} \le s^{12/25}, \qquad r^a \le s^{6/25}.$$

Interpolation needs $M(2\Lambda - 1) + 1 \le 2Cs^{26/25} \le s^4$ points, which the ring provides, and $M > H \ge s/\sqrt{U} \ge U \ge u$.

The packing inequalities. Using $M > s/\sqrt{U}$, (G2), (G3) and $2\sqrt{U} \le 2^{4U}$,

$$\begin{aligned} 2D_{\mathrm{arr}}^2\Delta \le 2^{96U+1}\Delta \le 2^{1-4U}s \le \frac{s}{\sqrt{U}} < M, \\ \Lambda D_{\mathrm{arr}} \le 2^{52U} \le \frac{s}{\sqrt{U}} < M, \qquad 2\Lambda \le 2^{4U+1} \le \Delta. \end{aligned} \tag{A.4}$$

For the middle inequality note $s \ge 2^{100U}\Delta \ge 2^{104U+1}$.

Lemma A.1 (Packing)**.** *Assume* (A.4)*. One can label the vertices on each side by elements of $[r]^a$ and place the edges into* packets *of Λ slots so that:*

1. *there are at most $3M/\Lambda$ packets, every edge occupies the same slot at its row end and at its column end, and unused slots carry no ports;*
2. *in every packet and every coordinate c the unordered pair of row and column symbols is fixed, and when the two symbols differ, bit c of the slot number records which of them is the row symbol;*
3. *for every coordinate and every symbol, at most $3M/r$ real ports on each side belong to vertices with that symbol in that coordinate.*

The construction takes time and space polynomial in s and the number of vertices.

Proof. The *pair label* of an edge is the pair (row label, column label), one of D_{arr} values; let X_γ be the number of edges with pair label γ. For independent uniform labels, using $\sum_\gamma X_\gamma = M$,

$$\mathbb{E}\sum_{\gamma}\Big(X_\gamma - \frac{M}{D_{\mathrm{arr}}}\Big)^2 = \sum_{e,e'}\Big(\Pr[\gamma(e) = \gamma(e')] - \frac{1}{D_{\mathrm{arr}}}\Big),$$

the sum running over ordered pairs of edges. If e and e' have disjoint endpoints their labels are independent and the term vanishes. At most $2M\Delta$ ordered pairs, $e = e'$ included, share an endpoint, and each term is at most one, so the expectation is at most $2M\Delta$. Labelling the vertices one at a time and choosing, among the r^a labels of the next vertex, one that

does not increase the conditional expectation [Rag88] gives a deterministic labelling with the same bound; each conditional expectation is a sum over pairs of edges of probabilities involving at most four vertices, so it is computed exactly in polynomial time. Every load is then at most $M/D_{\mathrm{arr}} + \sqrt{2M\Delta}$. Pad every pair-label class to H_{pad} slots, the largest load rounded up to a multiple of Λ; by (A.4), $\sqrt{2M\Delta} \le M/D_{\mathrm{arr}}$ and $\Lambda \le M/D_{\mathrm{arr}}$, so

$$H_{\mathrm{pad}} \le \frac{M}{D_{\mathrm{arr}}} + \sqrt{2M\Delta} + \Lambda \le \frac{3M}{D_{\mathrm{arr}}}.$$

Group the pair labels by the unordered pair of symbols in each coordinate. If a' coordinates of a group have unequal symbols, the group consists of $2^{a'}$ pair labels, which differ in orientation. Form packets of Λ slots by taking $2^{a-a'}$ slots of each of these $2^{a'}$ labels; this is possible since H_{pad} is divisible by Λ. Number the positions of a packet from 0 to $\Lambda - 1$ so that bit c gives the orientation when the pair in coordinate c is unequal and indexes copies otherwise, with one convention throughout a packet. There are $H_{\mathrm{pad}} D_{\mathrm{arr}}/\Lambda \le 3M/\Lambda$ packets. In a fixed coordinate a fixed symbol is the row symbol of exactly $1/r$ of the padded pair labels, so it has at most $H_{\mathrm{pad}} D_{\mathrm{arr}}/r \le 3M/r$ real row ports, and likewise for columns. □

A.3. **A Cauchy deformation.** Fix an active call, its packets and its lifted tensors. Put

$$S = \{-\Lambda, -\Lambda + 1, \dots, \Lambda - 1\}, \qquad \Phi(T) = \prod_{t \in S} (T - t), \qquad \alpha = 2^{a+1}.$$

For a row slot x and a column slot y of the same packet put

$$W_{xy}(T) = \frac{\Phi(T)}{\Phi'(0)\,(T - (y - x))}, \tag{A.5}$$

a polynomial of degree $2\Lambda - 1$ over $\mathbb{Q}$, since $y - x \in S$; entries between different packets are zero, and W is restricted to real ports. At $T = 0$ the diagonal entries are one and the others vanish, because $\Phi(T)/T \to \Phi'(0)$ and every other quotient keeps the factor T; so $W(0)$ is the identification of row and column ports given by the edges. Let $A(Y)$ be the product of the lifted row tensors on a row assignment $Y \in \{0,1\}^M$ and $B(X)$ the product of the lifted column tensors on a column assignment X. With $l_0(z) = 1 - z$ and $l_1(z) = z$ put

$$F(T) = \sum_{X,Y \in \{0,1\}^M} A(Y) B(X) \prod_{\text{row ports } x} l_{Y_x}\big((W(T)X)_x\big). \tag{A.6}$$

Its coefficients lie in $\mathbb{K}[V]/(V^{\ell})$, its degree is at most $M(2\Lambda - 1)$, and, since $\prod_x l_{Y_x}(X_x) = [Y = X]$, $F(0) = \sum_X A(X)B(X)$ is the contraction of the lifted tensors.

Lemma A.2 (Integrality)**.** *For every $t \in S$ the entries of $W(t + \alpha z)$ lie in $\mathcal{O}[z]$ and $F(t + \alpha z) \in \mathcal{A}[z]$. Given the coordinates of ζ modulo 2^p, each $W_{xy}(t + \alpha\zeta) \bmod 2^p$ is computed in time polynomial in s and p.*

Proof. S is a complete residue system modulo 2^{a+1}. For $q \ne t$ in S one has $v_2(t - q) \le a$, and the number of q with $v_2(t - q) = k$ does not depend on t; hence $v_2(\Phi'(t)) = v_2(\Phi'(0))$. In

$$\frac{\Phi(t + \alpha z)}{\Phi'(0)} = \frac{\alpha z \prod_{q \ne t} (t - q + \alpha z)}{\Phi'(0)}$$

each factor $t - q + \alpha z$ is divisible coefficientwise by $2^{v_2(t-q)}$, so after cancelling these powers against the denominator the polynomial is divisible by 2^{a+1}. Formula (A.5) omits the factor indexed by $q = t - (y - x)$. If $y - x = t$ the omitted factor is αz and the rest is integral; otherwise αz remains, the omitted factor has content $2^{v_2(t-y+x)}$ with $v_2(t - y + x) \le a$, and the quotient is still integral. Substituting in (A.6) gives the claim for F. To evaluate, divide each factor $t - q + \alpha z$ by $2^{v_2(t-q)}$, cancel the remaining powers of two against $\Phi'(0)$, invert its odd part modulo 2^p, and multiply the $2\Lambda - 1$ factors at ζ; recall $\Lambda \le s$. □

A.4. **Tapered Hermite recovery.**

Lemma A.3 (Recovery)**.** *Let $1 \le u \le a$ and $M \ge u$, and put $h = \lceil 3M/u \rceil$. For $t \in S \setminus \{0\}$ let*

$$i = v_2(t), \qquad q_t = q(i) = h\max(0, u - i), \qquad D_t = \Lambda/2^i.$$

Let $F \in (\mathbb{K}[V]/(V^\ell))[T]$ have degree at most $M(2\Lambda-1)$, with $F(0) \in \mathcal{A}$ and $F(t+\alpha z) \in \mathcal{A}[z]$ for all $t \in S$. Then $F(0) \bmod 2^p$ is determined by, and computable in polynomial time from, the first q_t coefficients of

$$F(t + \alpha z) \bmod 2^{p_t}, \qquad p_t = p + 8hD_t + 8M(a+1),$$

for the t with $q_t > 0$. In particular $p_t > p$.

Proof. There are Λ odd points in S, each with $q_t = hu \ge 3M$, so $\sum_{t\ne 0} q_t \ge 3M\Lambda > M(2\Lambda-1)$; also $q_t \le hu \le 3M + u \le 4M$. For t with $q_t > 0$ put

$$E_t(T) = \prod_{q \in S\setminus\{0,t\}} \Big(\frac{T-q}{t-q}\Big)^{q_q}.$$

We claim

$$F(0) = \sum_{t:\, q_t>0} E_t(0)\, \Big[\frac{F(t+\alpha z)}{E_t(t+\alpha z)}\Big]_{<q_t}\Big(-\frac{t}{\alpha}\Big), \tag{A.7}$$

where $[\cdot]_{<q}$ truncates a power series in z below degree q. Replace 0 by an indeterminate T and $-t/\alpha$ by $(T-t)/\alpha$ on the right. The summand of t agrees with F to order q_t at $T = t$ and vanishes to order $q_{t'}$ at every other node t', and its degree is below $\sum q_t$. So the difference with F is divisible by $\prod_{t\ne 0}(T-t)^{q_t}$ and has smaller degree, hence vanishes; this is Hermite interpolation by the Chinese remainder theorem [vzGG13, Section 5.6].

Each factor of $E_t(t+\alpha z)$ is a power of $1 + \alpha z/(t-q)$, which is integral with constant term one, so the reciprocal truncated below q_t is integral and the quotient in (A.7) loses no precision.

Fix $i = v_2(t)$ and $D = D_t$. Points q with $v_2(q) < i$ have $v_2(t-q) = v_2(q)$ and contribute nothing to $v_2(E_t(0))$. For $1 \le j \le a - i$ there are $D/2^j$ points with $v_2(q) = i + j$, for which $v_2(t-q) = i$, and $D/2^j$ points with $v_2(t-q) = i + j$, for which $v_2(q) = i$; these exhaust the points of valuation at least i other than t, since the nonzero differences have valuation at most a. Therefore

$$v_2(E_t(0)) = \sum_{q\ne 0,t} q_q(v_2(q) - v_2(t-q)) = D\sum_{j=1}^{a-i} \frac{j}{2^j}(q(i+j) - q(i)) \;\ge\; -6Dh, \tag{A.8}$$

because $q(i+j) - q(i) \ge -hj$ and $\sum_{j\ge 1} j^2/2^j = 6$. Evaluating a polynomial of degree below $q_t \le 4M$ at $-t/\alpha$, whose valuation is at least $-(a+1)$, loses at most $4M(a+1)$ powers of two. Hence replacing each quotient coefficient by any lift of its residue modulo 2^{p_t} changes the summand by an element of $2^p\mathcal{A}$, since $p_t - p > 6Dh + 4M(a+1)$. The multipliers $E_t(0)(-t/\alpha)^j$ are rationals of polynomial bit length, so the combination is carried out exactly with a common denominator, after which the powers of two are cancelled and the result reduced modulo 2^p. □

The tapering of the multiplicities is what makes the precision affordable. If every level below u received the same multiplicity $q_* = \Theta(M)$, the level $i = u - 1$ would lose $q_* D\sum_j j/2^j = \Theta(MD)$ powers of two in (A.8); with multiplicities decreasing by $h = \Theta(M/u)$ per level the loss is $O(MD/u)$. Figure 6 compares the two choices.

At a call of precision p apply Lemma A.3 with the parameters (A.1). For each t with $q_t > 0$ obtain the whole polynomial $F(t+\alpha z) \bmod 2^{p_t}$ by evaluating it at $M(2\Lambda-1)+1 < s^4$ of the interpolation points of Section A.1 and interpolating; the values $F(t+\alpha\zeta)$ are computed in Section A.5 by child calls of precision p_t.

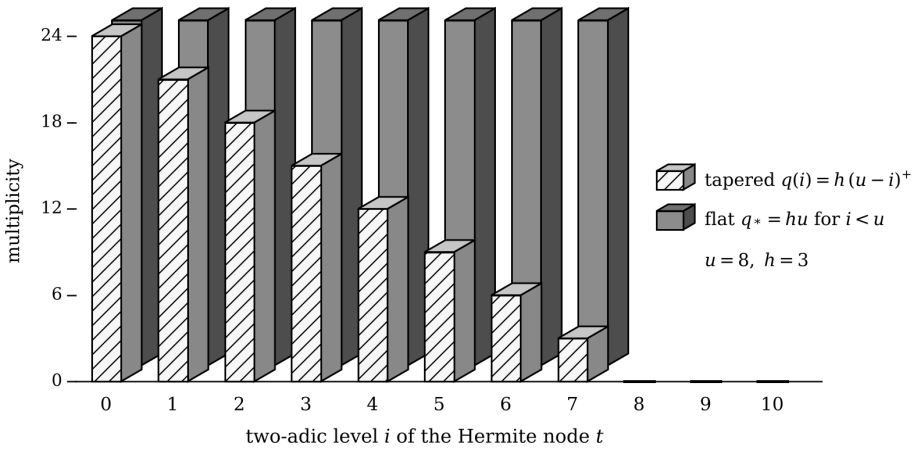


FIGURE 6. Multiplicities of the Hermite nodes by two-adic level, for $u = 8$ and $h = 3$. The tapered choice $q(i) = h \max(0, u - i)$ of Lemma A.3 (front) against a flat choice $q_* = hu$ on every level below u (back). Both give the odd points the multiplicity hu that the degree count needs; only the tapered one keeps the loss (A.8) at $O(Dh)$.

Precision along a path. Let π_d bound the precision of the calls at depth d, with $\pi_0 = P_0 \le Cs$. At an active call at depth d we have $h \le 4M/u$, $D_t \le \Lambda = 2^{a_d}$ and $a + 1 \le 2^a$, so

$$p_t - p = 8hD_t + 8M(a+1) \le \frac{32M2^{a_d}}{u} + 8M2^{a_d} \le 40\,Cs\,8^{-d}2^{a_d}.$$

Since $a_j \le a_{d-1}$ for $j < d$,

$$\pi_d \le Cs + 40Cs \sum_{j<d} 8^{-j}2^{a_j} \le \Big(1 + \frac{320}{7}\Big)Cs\,2^{a_{d-1}} \le 47\,Cs\,2^{a_{d-1}}. \tag{A.9}$$

In particular every precision is at most $47Cs\,2^{4U}$, polynomial in s.

The precision of a node. Let t have $i = v_2(t) < u$. Then $D_t = 2^{a_d - i} \ge 2^{a_d - u + 1} = 2^{a_{d-1} + u + 1}$. We bound the three terms of p_t. First, $8hD_t \le 32MD_t/u$. Second, $8M(a+1) \le MD_t/u$, because $8u(a+1) \le 4a(a+1) \le 2^{a/2+1} \le 2^{a-u+1}$, using $u \le a/2$ and $a \ge 32$. Third, $p \le MD_t/u$: by (A.9) and $M > s/\sqrt{U}$ it suffices that $47C\sqrt{U}\,u \le 2^{u+1}$, and by (A.3) it suffices that $47C^{6/5}(u+1)^{8/5} \le 2^{u+1}$, which holds for $u \ge 16 + 2\log C$. Hence

$$p_t \le 34\,\frac{MD_t}{u}. \tag{A.10}$$

A.5. **Modes for interactions between groups.** Fix t with $i = v_2(t) < u$, put $D = \Lambda/2^i$, fix an interpolation point ζ, and work modulo 2^{p_t} with $T = t + \alpha\zeta$. Since $i < u \le a$, coordinate i exists. Group the vertices on both sides by their symbol in coordinate i; there are r groups. Entries of $W(T)$ whose two ports lie in the same group are *local*.

The matrix $W(t)$ is local. Indeed $W_{xy}(t) = 0$ unless $y = x + t$, and adding t, whose lowest set bit is bit i, flips bit i of the slot number without a carry from below. When the pair of symbols of the packet in coordinate i is unequal, bit i of a slot records the orientation, so a row slot and a column slot lie in the same group exactly when their bits i differ; when the pair is equal, the whole packet is one group.

In a packet with unequal pair, write the column slots as $y = y_0 + 2^{i+1}j$ with $0 \le y_0 < 2^{i+1}$; the group of y depends only on y_0, so these *tiles* are monochromatic. If a row x and the tile of y_0 lie in different groups, then $v_2(t - y_0 + x) \le i$, for otherwise $y_0 \equiv x + t \bmod 2^{i+1}$ and their bits i would differ. The same holds with T in place of t, since $v_2(\alpha\zeta) > i$. Expanding

geometrically,

$$W_{xy}(T) \equiv \sum_{\substack{\nu\ge 0\\ \nu+1<p_t}} \frac{\Phi(T)}{\Phi'(0)(T-y_0+x)}\Big(\frac{2^{i+1}}{T-y_0+x}\Big)^{\nu} j^{\nu} \pmod{2^{p_t}}. \tag{A.11}$$

The first factor has valuation at least $a+1-i\ge 1$ by the proof of Lemma A.2, and the ratio has valuation at least one, so the term of index ν is divisible by $2^{\nu+1}$ and the remainder of the geometric series by 2^{p_t}. For each packet, tile and ν introduce a *mode* Z of weight $w_Z=\nu+1$, standing for

$$L_Z(X) = \sum_{\text{real columns } y \text{ in the tile}} j^{\nu} X_y \qquad (j^0=1).$$

Its *source* is the group of the tile and its *target* the other group of the packet's pair. By (A.11), each row form $(W(T)X)_x$ is its local part plus a combination of the modes targeted at the group of x, with coefficients divisible by 2^{w_Z}. A packet has 2^{i+1} tiles, so there are at most

$$\frac{3M}{\Lambda}\,2^{i+1} = \frac{6M}{D} \tag{A.12}$$

modes of each weight.

Replace $W(T)$ in (A.6) by the truncated series (A.11); since $A(Y), B(X)\in\mathcal{A}$ and every row form is integral, this changes $F(T)$ by an element of $2^{p_t}\mathcal{A}$. Expand the product as a polynomial in the modes. The coefficient of a monomial Z^{β} is divisible by $2^{\sum_Z w_Z\beta_Z}$ and the values $\prod_Z L_Z(X)^{\beta_Z}$ are integers, so modulo 2^{p_t} only the exponent vectors with

$$\sum_Z w_Z\beta_Z < p_t \tag{A.13}$$

contribute. Their number is at most $\exp(64M/\sqrt{u})$. Indeed, for $\theta>0$ the generating function bounds the number by

$$\exp\Big(\theta p_t + \frac{6M}{D}\sum_{j\ge1} -\ln(1-e^{-\theta j})\Big), \qquad \sum_{j\ge1} -\ln(1-e^{-\theta j}) = \sum_{k\ge1}\frac{1}{k(e^{\theta k}-1)} \le \frac{\pi^2}{6\theta},$$

and $\theta=\sqrt{M/(Dp_t)}$ gives $\exp((1+\pi^2)\sqrt{Mp_t/D})$; then (A.10) gives $\sqrt{Mp_t/D}\le\sqrt{34}\,M/\sqrt{u}$, and $(1+\pi^2)\sqrt{34}<64$. The vectors are enumerated with polynomial delay by choosing exponents one variable at a time within the remaining budget.

A.6. **The child networks.** For one vector β satisfying (A.13) and one group $g\in[r]$, the child network $\mathcal{N}_{g,\beta}$ is built as follows. On one side put all current vertices of group g, from both sides, with their ports and their lifted entries reduced modulo 2^{p_t}. On the other side put one vertex for each packet that has ports of group g, joined to every port of group g in that packet; its ports are the row bits Y and column bits X of group g in the packet. Its tensor is the coefficient of $\prod_Z Z^{\beta_Z}$, over the modes Z of the packet targeted at g, in the product of the row factors l_{Y_x} of group g in the packet, written in the local column bits and the formal modes, multiplied by $\prod_Z L_Z(X)^{\beta_Z}$ over the modes of the packet whose source is g. Empty products are one; a positive exponent on a mode with no target rows, or on a mode whose tile has no real column, gives zero. A packet with no port of group g would contribute the factor one and is omitted.

Every mode has one source and one target, the coefficient extraction factors over the groups of the rows, and the substitution of each mode uses only bits of its source group. Distributing the product in (A.6) and summing the bits of each group independently therefore gives

$$F(t+\alpha\zeta) \equiv \sum_{\beta}\prod_{g=1}^{r} Z(\mathcal{N}_{g,\beta}) \pmod{2^{p_t}}, \tag{A.14}$$

the sum over the vectors satisfying (A.13); the omitted monomials vanish modulo 2^{p_t}.

By Lemma A.1(3) a child has at most $6M/r = 3M/32 < M/8$ edges. Old vertices keep their degrees, and a packet vertex has at most $2\Lambda \le \Delta$ ports by (A.4), so the invariants of Section A.2 hold inductively. A child has at most $3M/\Lambda$ new vertices, so along a path the number of vertices stays below $s^b + 3Cs$. A packet entry is a coefficient of a product of at most Λ affine factors, each with polynomially many terms, and is computed in time $s^{O(\Lambda)} = 2^{O(s^{1/25}\log s)}$ and polynomial space; powers $L_Z(X)^{\beta_Z}$ are computed by repeated squaring. A new tensor is described by its packet data, β, ζ and its creation precision, and old tensors keep their descriptions, shared by reference, so entry routines do not compose along the recursion.

A.7. **Proof of Theorem 8.4.** Embed the entries in $\mathcal{A}_{P_0}$ and run the recursion. At an active call, for every t with $q_t > 0$ and every interpolation point, compute (A.14) from the child contractions, interpolate $F(t + \alpha z)$, and apply Lemma A.3; at a leaf sum directly. Correctness follows by induction from Lemmas A.2 and A.3 and (A.14).

Number of calls. An active call at depth d makes at most

$$\underbrace{2\Lambda}_{t} \cdot \underbrace{2M\Lambda}_{\zeta} \cdot \underbrace{64}_{g} \cdot e^{64M/\sqrt{u}} \;\le\; 256\, Cs\, 2^{8U}\, e^{64M/\sqrt{u}}$$

child calls. By (A.3) and $M \le Cs8^{-d}$,

$$\frac{64M}{\sqrt{u_d}} \le 64\sqrt{\tfrac{17}{16}}\,\frac{Cs}{\sqrt{U}}\Big(\frac{\sqrt{2}}{8}\Big)^d, \qquad \sum_{d\ge 0}\frac{64M_d}{\sqrt{u_d}} \le \frac{64\sqrt{17/16}}{1-\sqrt{2}/8}\cdot\frac{Cs}{\sqrt{U}} < 81\,\frac{Cs}{\sqrt{U}},$$

where M_d is the size of the call at depth d on a path. Active depths satisfy $d \le d_{\max} < \frac{1}{3}\log(C\sqrt{U})$ by (A.2). The number of calls is therefore at most

$$(d_{\max}+2)\,(256Cs^{27/25})^{d_{\max}+1}\, e^{81Cs/\sqrt{U}} \le 2^{(117C+1)s/\sqrt{U}},$$

using $81/\ln 2 < 117$ and

$$(2 + \tfrac{1}{3}\log(C\sqrt{U}))\log(256Cs^{27/25}) \le \frac{s}{\sqrt{U}}. \tag{A.15}$$

For (A.15): (G1) gives $C \le U^{5/2}$, so the left side is at most $2U\log s \le (\log s)^2/50$, while the right side is at least $s/\log s$.

Leaves and total. A leaf sums $2^M \le 2^H \le 2^{2s/\sqrt{U}+1}$ assignments, each a product of at most $s^b + 3Cs$ entries; an entry costs at most T_{ent} for an original tensor and at most $2^{O(s^{1/25}\log s)}$ for a packet tensor, and arithmetic is polynomial in s and the precision. All other work at an active call is polynomial per child. The total time is at most

$$(s+L)^c\, T_{\text{ent}}\, 2^{(117C+1)s/\sqrt{U}} \cdot 2^{2s/\sqrt{U}+1} \cdot 2^{O(s^{1/25}\log s)} \le (s+L)^c\, T_{\text{ent}}\, 2^{(117C+4)s/\sqrt{U}}.$$

For space, traverse the recursion depth first, keeping at each of the at most $d_{\max}+2$ levels the Taylor data of the current call, its indices and its new tensor descriptions, all of polynomial size by (A.9). Finally read off the coordinates of the result in $B_{P_0,\ell}$. □

The thresholds are large. Corollary 8.5 starts at $\log s_0 = 128(U_0(C)+1)/\delta'$, which for $C = 22$ is $s_0 = 2^{22144/\delta'}$, and the bound $2^{(117C+4)s/\sqrt{U}}$ is below the trivial 2^{Cs} only when $\sqrt{U} > 117 + 4/C$. The constants were chosen for a short proof and are not optimised; the statements are asymptotic.

Appendix B. Constants and calculations

This appendix collects the parameters of Appendix A and of Sections 9 and 11, and the elementary inequalities behind them, so that each can be checked in one place.

B.1. **Parameters.** Table 3 lists the parameters of an active call at depth d and the bounds that the proof of Theorem 8.4 uses for them. Throughout, $C \geq 1$ is the constant of the edge bound, U the number of two-adic levels at the root and Δ the degree bound, subject to (G1)–(G3).

TABLE 3. Parameters of the recursion and the bounds used for them.

Quantity	Definition	Bound used
leaf threshold H	$\lceil s/\lfloor\sqrt{U}\rfloor\rceil$	$s/\sqrt{U} \leq H \leq 2s/\sqrt{U}+1$
groups r	64	child size $\leq 6M/r < M/8$
levels u_d	$\lfloor U/2^d \rfloor$	$u_d \geq 16 + 2\log C$, $u_d \geq \frac{16}{17}U2^{-d}$
exponent a_d	$2\sum_{j\leq d} u_j$	$2u_d \leq a_d \leq 4U$, $a_d \geq 32$, even
packet length Λ	2^a	$\Lambda \leq 2^{4U} \leq s^{1/25}$
pair labels D_{arr}	$r^{2a} = 2^{12a}$	$D_{\mathrm{arr}} \leq 2^{48U} \leq s^{12/25}$
depth d	on an active path	$8^d < C\sqrt{U}$
multiplicity step h	$\lceil 3M/u \rceil$	$h \leq 4M/u$
node precision p_t	$p + 8hD_t + 8M(a+1)$	$p_t \leq 34MD_t/u$
path precision π_d		$\pi_d \leq 47Cs\,2^{a_{d-1}}$
modes per weight		$\leq 6M/D$
mode monomials	$\sum w_Z\beta_Z < p_t$	$\leq \exp(64M/\sqrt{u})$
children of a call		$\leq 256Cs\,2^{8U}e^{64M/\sqrt{u}}$

B.2. **Elementary inequalities.** The following facts are used in Appendix A; each is checked by machine as indicated in Appendix C.

(E1) $U_0(1) = 30$ and $U_0(22) = 172$.
(E2) $2\sqrt{U} \leq 2^{4U}$ for every integer $U \geq 1$, used in (A.4).
(E3) $4a(a+1) \leq 2^{a/2+1}$ for every even $a \geq 20$, used for the second term of (A.10).
(E4) $47C^{6/5}(u+1)^{8/5} \leq 2^{u+1}$ for $C \geq 1$ and $u \geq 16 + 2\log C$, used for the third term of (A.10). Writing $C = 2^\kappa$, it reads $\log 47 + \frac{6}{5}\kappa + \frac{8}{5}\log(u+1) \leq u+1$. The difference of the two sides increases with u for $u \geq 2$, since $\frac{d}{du}(u - \frac{8}{5}\log(u+1)) = 1 - \frac{8}{5(u+1)\ln 2} > 0$; at $u = 16 + 2\kappa$ it equals $17 - \log 47 + \frac{4}{5}\kappa - \frac{8}{5}\log(17+2\kappa)$, which is $4.9\ldots$ at $\kappa = 0$ and increases with κ.
(E5) $1 + \frac{320}{7} \leq 47$, used in (A.9).
(E6) $(1+\pi^2)\sqrt{34} < 64$, used for the number of mode monomials; indeed $(1+\pi^2)\sqrt{34} < 10.870 \times 5.831 < 63.4$.
(E7) $64\sqrt{17/16}/(1-\sqrt{2}/8) < 81$; the left side is $80.14\ldots$.
(E8) $81/\ln 2 < 117$; the left side is $116.86\ldots$.
(E9) $\sum_{j\geq 1} j^2/2^j = 6$, used in (A.8).
(E10) $6/64 < 1/8$, the child size.
(E11) $16(117C+4) = 1872C + 64$ and $(117C+4)\sqrt{104} < 1194C + 41$ for $C \geq 1$.
(E12) $(1872 \cdot 22 + 65)\sqrt{3} < 71500$, $71500\sqrt{2} < 101200$ and $4/(5 \cdot 10^6) < 1/(12 \cdot 101200)$, used in Theorems 9.1 and 11.6.

B.3. **The conditions of Corollary 8.5.** With $x = \log s$, $\delta \leq \frac{1}{2}$, $U = \lfloor \delta\lceil x \rceil/128 \rfloor$ and $\delta x \geq 128(U_0(C)+1)$, the three conditions of Theorem 8.4 reduce to linear inequalities in x:

$$\text{(G2)}: \quad \frac{25\delta}{32}(x+1) + (1-\delta)x + 1 \leq x, \qquad \text{(G3)}: \quad \frac{\delta(x+1)}{32} + 1 \leq (1-\delta)x,$$

both of which follow from $\delta x \geq 128$, and (G1) is $\delta x/128 - 1 \geq U_0(C)$. Table 4 gives the resulting thresholds.

TABLE 4. Thresholds and constants. The size s is measured as $\log s$.

Statement	Range of validity	Constant
Theorem 8.4	(G1)–(G3)	exponent $(117C+4)s/\sqrt{U}$
Corollary 8.5	$\log s \geq 128(U_0(C)+1)/\delta'$	$(1872C+64+A)/\sqrt{\delta'}$
Corollary 8.6	$\log(s/\Delta) \geq 100U_0(C)$	$1194C+41$
Lemma 9.3	$\log s \geq 40/\delta$	degree $\leq 3s^{1-\delta/2}$
Theorem 9.1	$\log s \geq 66432/\delta$	$A_0(\delta) = 71500/\sqrt{\delta}$
Theorem 11.6	infinitely many N	$c(\delta) = \sqrt{\delta'}/(5\cdot 10^6)$

B.4. **Changes from the original parameters.** The proof of OpenAI uses $\Delta = \lceil s^{9/10} \rceil$, $\varepsilon = 2^{-17}$, $u_d = \lfloor \varepsilon\lambda/2^d \rfloor$ and the leaf threshold $\lceil s/\lfloor\sqrt{\lambda}\rfloor\rceil$. In the notation above this is $U = \lfloor 2^{-17}\lambda \rfloor$, and since $96\cdot 2^{-17} < 2^{-10}$ the inequality (A.4) holds for every degree up to $s^{1-1/1024}$ with these values. Table 5 lists the inequalities whose form changes when the degree becomes $s^{1-\delta}$ and $\varepsilon = \delta/128$.

TABLE 5. Inequalities of the proof at degree $s^{9/10}$ and at degree $s^{1-\delta}$, $\delta \leq \frac{1}{2}$, $U = \lfloor \delta\lambda/128 \rfloor$.

Inequality	Degree $s^{9/10}$	Degree $s^{1-\delta}$
levels	$u \geq \lambda^{4/5}$	$u \geq 16 + 2\log C$
packet length	$\Lambda \leq s^{1/1000}$	$\Lambda \leq (2s)^{\delta/32}$
pair labels	$D_{\mathrm{arr}} \leq s^{1/1000}$	$D_{\mathrm{arr}} \leq (2s)^{3\delta/8}$
$2D_{\mathrm{arr}}^2\Delta$	$O(s^{9/10+1/500})$	$\leq 8s^{1-\delta/4}$
path precision	$s\lambda^{O(1)}2^{a_{d-1}}$	$47Cs\,2^{a_{d-1}}$
node precision	$O_C(MD_t/u)$	$34MD_t/u$
packet entry	$\exp(O(\Lambda\log s))$	$\exp(O(s^{\delta/32}\log s))$
calls	$\exp(O_C(s/\sqrt{\lambda}))$	$\exp(81Cs/\sqrt{U})$

B.5. **The chain of exponents.** The lower bound at fan-in $N^{1-\delta}$ is reached through four exponents, each a fixed fraction of the previous one:

$$\underbrace{N^{1-\delta}}_{\text{lower bound}} \impliedby \underbrace{s^{1-\delta/2}}_{\text{counting}} \impliedby \underbrace{s^{1-\delta/6}}_{\text{network degree}} \impliedby \underbrace{U = \lfloor \delta\lambda/768 \rfloor}_{\text{two-adic levels}}.$$

The first step loses a factor two in δ because the test circuit of Theorem 11.5 on $N = dn$ inputs is only known to have size at least n; the second loses a factor three because the roots of big gates have degree about $(s/K)\beta'$ (Remark 9.5); the third is the choice $\varepsilon = \delta/128$ of Corollary 8.5. Each loss changes only the constant $c(\delta)$.

APPENDIX C. PROGRAMS THAT CHECK THE FINITE STATEMENTS

Two programs accompany the paper. They check finite statements and constants; the proofs in the text do not depend on them.

C.1. **Lean.** A Lean 4 project (Lean v4.35.0 with Mathlib v4.35.0) formalises the combinatorial core of the counting network and the arithmetic of the proofs. It builds without unproved statements and without added axioms; its audit file lists 46 results, each depending only on a subset of the three standard axioms of Lean (propositional extensionality, choice and quotient soundness). Table 6 lists its files.

TABLE 6. The Lean project.

File	Content	Results
`Circuit`	a circuit as a constraint per gate; the contraction with a pin equals the number of satisfying inputs (Lemma 9.2, in the form with one bit per signal); products of equalities along a path of an equality tree	9
`Blocks`	block range $0 \le P + K2^{\beta} < 2^{\beta'}$; partial sums; decoding at the root; the edge count $6W + 6 \le 11s$ and $22s$; block edges $\lceil f/K \rceil \beta' \le 2f$	7
`GT`	the matrix $[x \ge y]$ on a finite order has determinant one and full rank over every field	5
`Valuation`	identity (A.8) and the bound $\ge -6Dh$, the equality $v_2(\Phi'(t)) = v_2(\Phi'(0))$, integrality of $W(t + \alpha z)$ and $W(0) = I$, all for small a by kernel computation; the computable valuation equals Mathlib's	7
`Transfer`	$3A_1\eta < \frac{1}{4}$; membership of the test circuit, $(dn)^{1-\delta_2} \le n^{1-\delta_1}$ for $n \ge d^{(1-\delta_2)/(\delta_2-\delta_1)}$	2
`Constants`	(E2), (E3), (E5)–(E8), (E10)–(E12) and the conditions (G2), (G3) of Corollary 8.5; the constants of Theorem 12.4 for the cube	16

C.2. **Julia.** A Julia script, using only the standard library and exact integer and rational arithmetic, with 512-bit floating point only where π, square roots or logarithms occur, runs 44 checks in a few minutes, all of which pass. Table 7 lists them.

TABLE 7. The Julia checks.

Part	Statement	Checks
A	the contraction of $\mathcal{N}(C)$ equals $\#C$ on 120 random circuits with all gate kinds, big weighted gates split into blocks	2
B	occurrences, equality trees, block edges and the bound $22s$ on random circuits	2
C	Corollary 10.1: counts, binary search and self-reduction on 60 random programs	1
D	$v_2(\Phi'(t)) = v_2(\Phi'(0))$; identity (A.8) in 10328 cases; the bound $-6Dh$	3
E	$W(0) = I$; integrality of $W(t + \alpha z)$ for $a \le 4$	2
F	Hermite recovery (A.7) on random polynomials	1
G	properties (1)–(3) of Lemma A.1 and the load bound on random bipartite graphs, three parameter sets	15
H	(E1)–(E12); conditions (G1)–(G3) of Corollary 8.5 for $\delta \in \{0.01, \ldots, 0.50\}$; the degrees of Lemma 9.3 for six values of δ; the membership threshold of Theorem 11.5	13
I	Theorem 12.4: I_G equals its exact-gate form on all inputs for Q_2, Q_3; $\mathrm{rk}_A(I_G) \ge 2^{\lvert M \rvert}$ on 654 partitions; Lemma 12.3 on 25000 sets; the isoperimetric inequality on all subsets of Q_4; the constants	5

Acknowledgements

The contraction theorem on which Theorem A rests is due to OpenAI [OAI26]; Appendix A follows its proof and makes the degree and the number of two-adic levels into parameters. The lower bound uses the algorithmic method of Williams [Wil13, Wil14], the local reductions of Jahanjou, Miles and Viola [JMV18] and the hierarchy theorem of Žák [Zak83].

Author contributions

Deep Bhattacharjee is the sole author. He posed the problem in the form treated here, carried out the research and wrote the paper: the survey of Part I with its catalogue, the parametrised contraction theorem and its proof, the counting theorem, the lower bounds and their corollaries, the analysis of the limits of the method, the Lean 4 project of Table 6, the Julia programs of Table 7 and the figures.

Open Gluon and Anaxia

Two tools built by D. B. stand behind the way this project was organised. Open Gluon is a workspace that keeps pinned copies of the mathematical libraries, the source record of the project and its checks in one reproducible place. Anaxia, the engine of the eBLLM project, stores each construction as a route with its input, its output, the obligations that must be discharged and a test that makes the route fail when an obligation is skipped, and it certifies finite statements in exact rational arithmetic. Two of its constructions were used here. The first carries every constant of the contraction theorem through the recursion as an explicit obligation; it located the single inequality, the balance of the labelling, at which the degree bound $s^{9/10}$ enters, and so led to the parametrised form of Theorem 8.4 and to the analysis of Section 12.2. The second is exact certification: the Julia programs recompute the constants and thresholds of Table 4 in exact rational arithmetic. The engine proposes routes and tests them; every statement of this paper is proved in the text.

Declarations

Data and code availability. No data were generated. The paper, the Lean 4 project and the Julia programs are archived at doi:10.5281/zenodo.24163297. The Lean project builds over Mathlib without unproved statements and without added axioms, and every named result depends only on the three standard axioms of Lean. The catalogue of arXiv papers of Section 7, the program that produced it and the list of claims taken from Woeginger's page are archived with the first version of this paper, doi:10.5281/zenodo.23934035.

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Ethics approval. Not applicable. The work involves no human participants, animals or personal data.

References

[Aar16] S. Aaronson, $\mathsf{P} \stackrel{?}{=} \mathsf{NP}$, in: Open Problems in Mathematics, J. F. Nash, Jr. and M. Th. Rassias (eds.), Springer, Cham, 2016, pp. 1–122. doi:10.1007/978-3-319-32162-2_1

[AB09] S. Arora and B. Barak, *Computational Complexity: A Modern Approach*, Cambridge University Press, Cambridge, 2009. doi:10.1017/CBO9780511804090

[AB87] N. Alon and R. B. Boppana, *The monotone circuit complexity of boolean functions*, Combinatorica **7** (1987), no. 1, 1–22. doi:10.1007/BF02579196

[Ajt83] M. Ajtai, Σ^1_1-*formulae on finite structures*, Ann. Pure Appl. Logic **24** (1983), no. 1, 1–48. doi:10.1016/0168-0072(83)90038-6

[AK10] E. Allender and M. Koucký, *Amplifying lower bounds by means of self-reducibility*, J. ACM **57** (2010), no. 3, 1–36. doi:10.1145/1706591.1706594

[AKS04] M. Agrawal, N. Kayal and N. Saxena, *PRIMES is in P*, Ann. of Math. (2) **160** (2004), no. 2, 781–793. doi:10.4007/annals.2004.160.781

[ALMSS98] S. Arora, C. Lund, R. Motwani, M. Sudan and M. Szegedy, *Proof verification and the hardness of approximation problems*, J. ACM **45** (1998), no. 3, 501–555. doi:10.1145/278298.278306

[AS98] S. Arora and S. Safra, *Probabilistic checking of proofs*, J. ACM **45** (1998), no. 1, 70–122. doi:10.1145/273865.273901

[AW09] S. Aaronson and A. Wigderson, *Algebrization*, ACM Trans. Comput. Theory **1** (2009), no. 1, 1–54. doi:10.1145/1490270.1490272

[BGS75] T. Baker, J. Gill and R. Solovay, *Relativizations of the $\mathcal{P}$ =?$\mathcal{NP}$ question*, SIAM J. Comput. **4** (1975), no. 4, 431–442. doi:10.1137/0204037

[Bha26c] D. Bhattacharjee and U. Bhattacharya, *Iterated nerve complexes for k-fold sensor coverage*, Preprints (2026), version 2, 2 July 2026. doi:10.20944/preprints202606.1351.v2

[Bha26n] D. Bhattacharjee, *Neuromorphic Grothendieck topoi for grid-embedded circuit architectures*, SSRN preprint (2026). doi:10.2139/ssrn.6194939

[Bha26q] S. Singha Roy, D. Bhattacharjee, P. Samal, P. Nandi, S. N. Thakur, R. Sadhu, S. Bhattacharya, R. Dey, O. Frederick, R. Ghora and R. Patra, *Quantum algorithms on symplectic state manifolds*, SSRN preprint (2026). doi:10.2139/ssrn.6114107

[Bha26v1] D. Bhattacharjee, *Superlinear wire lower bounds toward P versus NP*, Preprints (2026), version 1, 5 November 2026. doi:10.20944/preprints202611.0576.v1

[BhaHex26] D. Bhattacharjee, *Subexponential weighted counting and exact optimisation*, Hexagon preprint hexagon:2610.00594v1, 29 October 2026, `https://hexagonmath.org/2610.00594v1`.

[BIP19] P. Bürgisser, C. Ikenmeyer and G. Panova, *No occurrence obstructions in geometric complexity theory*, J. Amer. Math. Soc. **32** (2019), no. 1, 163–193. doi:10.1090/jams/908

[Blu84] N. Blum, *A Boolean function requiring 3n network size*, Theoret. Comput. Sci. **28** (1984), no. 3, 337–345. doi:10.1016/0304-3975(83)90029-4

[BSW01] E. Ben-Sasson and A. Wigderson, *Short proofs are narrow—resolution made simple*, J. ACM **48** (2001), no. 2, 149–169. doi:10.1145/375827.375835

[Cai07] J.-Y. Cai, $\mathrm{S}_2^p \subseteq \mathrm{ZPP}^{\mathrm{NP}}$, J. Comput. System Sci. **73** (2007), no. 1, 25–35. doi:10.1016/j.jcss.2003.07.015

[Che11] D. Yu. Cherukhin, *Lower bounds for complexity of Boolean circuits of finite depth with arbitrary elements*, Discrete Math. Appl. **21** (2011), no. 4, 499–508. doi:10.1515/dma.2011.031

[CHLR26] L. Chen, S. Hirahara, Z. Li and H. Ren, *Symmetric Exponential Time Requires Near-Maximum Circuit Size*, J. ACM **73** (2026), no. 1, 1–21. doi:10.1145/3778166

[CHOPRS22] L. Chen, S. Hirahara, I. C. Oliveira, J. Pich, N. Rajgopal and R. Santhanam, *Beyond Natural Proofs: Hardness Magnification and Locality*, J. ACM **69** (2022), no. 4, 1–49. doi:10.1145/3538391

[CHR24] L. Chen, S. Hirahara and H. Ren, *Symmetric Exponential Time Requires Near-Maximum Circuit Size*, in: Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC 2024), 2024, pp. 1990–1999. doi:10.1145/3618260.3649624

[CIP06] C. Calabro, R. Impagliazzo and R. Paturi, *A duality between clause width and clause density for SAT*, in: 21st Annual IEEE Conference on Computational Complexity (CCC 2006), 2006, pp. 252–260. doi:10.1109/CCC.2006.6

[CIP09] C. Calabro, R. Impagliazzo and R. Paturi, *The Complexity of Satisfiability of Small Depth Circuits*, in: Parameterized and Exact Computation (IWPEC 2009), Lecture Notes in Comput. Sci. **5917**, Springer, Berlin, 2009, pp. 75–85. doi:10.1007/978-3-642-11269-0_6

[CKKSZ15] R. Chen, V. Kabanets, A. Kolokolova, R. Shaltiel and D. Zuckerman, *Mining Circuit Lower Bound Proofs for Meta-Algorithms*, Comput. Complexity **24** (2015), no. 2, 333–392. doi:10.1007/s00037-015-0100-0

[Clay] Clay Mathematics Institute, *P vs NP*, Millennium Prize Problems, official problem description by S. Cook, `https://www.claymath.org/millennium/p-vs-np/`.

[Coo71] S. A. Cook, *The complexity of theorem-proving procedures*, in: Proceedings of the Third Annual ACM Symposium on Theory of Computing (STOC 1971), 1971, pp. 151–158. doi:10.1145/800157.805047

[Coo73] S. A. Cook, *A hierarchy for nondeterministic time complexity*, J. Comput. System Sci. **7** (1973), no. 4, 343–353. doi:10.1016/S0022-0000(73)80028-5

[CR79] S. A. Cook and R. A. Reckhow, *The relative efficiency of propositional proof systems*, J. Symbolic Logic **44** (1979), no. 1, 36–50. doi:10.2307/2273702

[CSS18] R. Chen, R. Santhanam and S. Srinivasan, *Average-case lower bounds and satisfiability algorithms for small threshold circuits*, Theory Comput. **14** (2018), Art. 9, 1–55. doi:10.4086/toc.2018.v014a009

[CTW26] L. Chen, A. Tal and Y. Wang, *Superquadratic Lower Bounds for Depth-2 Linear Threshold Circuits*, in: Proceedings of the 58th Annual ACM Symposium on Theory of Computing (STOC 2026), 2026, pp. 874–885. doi:10.1145/3798129.3800802

[Din07] I. Dinur, *The PCP theorem by gap amplification*, J. ACM **54** (2007), no. 3, Art. 12. doi:10.1145/1236457.1236459

[ECCC] Electronic Colloquium on Computational Complexity, `https://eccc.weizmann.ac.il/`.

[Edm65] J. Edmonds, *Paths, Trees, and Flowers*, Canad. J. Math. **17** (1965), 449–467. doi:10.4153/CJM-1965-045-4

[FGHK23] M. G. Find, A. Golovnev, E. A. Hirsch and A. S. Kulikov, *Improving* $3N$ *circuit complexity lower bounds*, Comput. Complexity **32** (2023), no. 2, Art. 13. doi:10.1007/s00037-023-00246-9

[FLMV05] L. Fortnow, R. Lipton, D. van Melkebeek and A. Viglas, *Time-space lower bounds for satisfiability*, J. ACM **52** (2005), no. 6, 835–865. doi:10.1145/1101821.1101822

[For09] L. Fortnow, *The status of the P versus NP problem*, Commun. ACM **52** (2009), no. 9, 78–86. doi:10.1145/1562164.1562186

[FSS84] M. Furst, J. B. Saxe and M. Sipser, *Parity, circuits, and the polynomial-time hierarchy*, Math. Systems Theory **17** (1984), no. 1, 13–27. doi:10.1007/BF01744431

[Gas19] W. I. Gasarch, *Guest column: the third* $P =?$ *NP poll*, ACM SIGACT News **50** (2019), no. 1, 38–59. doi:10.1145/3319627.3319636

[GHKK18] A. Golovnev, E. A. Hirsch, A. Knop and A. S. Kulikov, *On the limits of gate elimination*, J. Comput. System Sci. **96** (2018), 107–119. doi:10.1016/j.jcss.2018.04.005

[GJ79] M. R. Garey and D. S. Johnson, *Computers and Intractability: A Guide to the Theory of NP-Completeness*, W. H. Freeman, San Francisco, 1979.

[GKST18] A. Golovnev, A. S. Kulikov, A. V. Smal and S. Tamaki, *Gate elimination: circuit size lower bounds and #SAT upper bounds*, Theoret. Comput. Sci. **719** (2018), 46–63. doi:10.1016/j.tcs.2017.11.008

[Hak85] A. Haken, *The intractability of resolution*, Theoret. Comput. Sci. **39** (1985), 297–308. doi:10.1016/0304-3975(85)90144-6

[Has86] J. Håstad, *Almost optimal lower bounds for small depth circuits*, in: Proceedings of the 18th Annual ACM Symposium on Theory of Computing (STOC 1986), 1986, pp. 6–20. doi:10.1145/12130.12132

[Has94] J. Håstad, *On the size of weights for threshold gates*, SIAM J. Discrete Math. **7** (1994), no. 3, 484–492. doi:10.1137/S0895480192235878

[Has98] J. Håstad, *The shrinkage exponent of de Morgan formulas is* 2, SIAM J. Comput. **27** (1998), no. 1, 48–64. doi:10.1137/S0097539794261556

[Her14] T. Hertli, *3-SAT faster and simpler—Unique-SAT bounds for PPSZ hold in general*, SIAM J. Comput. **43** (2014), no. 2, 718–729. doi:10.1137/120868177

[HI25] S. Hirahara and R. Ilango, *NP-hardness of the Minimum Circuit Size Problem from Well-Studied Assumptions*, in: 66th Annual IEEE Symposium on Foundations of Computer Science (FOCS 2025), 2025, pp. 1648–1664. doi:10.1109/FOCS63196.2025.00087

[Hir22] S. Hirahara, *NP-hardness of learning programs and partial MCSP*, in: 63rd Annual IEEE Symposium on Foundations of Computer Science (FOCS 2022), 2022, pp. 968–979. doi:10.1109/FOCS54457.2022.00095

[Hir23] S. Hirahara, *Non-Black-Box Worst-Case to Average-Case Reductions within* NP, SIAM J. Comput. **52** (2023), no. 6, FOCS18-349–FOCS18-382. doi:10.1137/19M124705X

[HIR25] Y. Huang, R. Ilango and H. Ren, *NP-Hardness of Approximating Meta-Complexity: A Cryptographic Approach*, SIAM J. Comput. **54** (2025), no. 4, 819–886. doi:10.1137/23M1608483

[HMPST93] A. Hajnal, W. Maass, P. Pudlák, M. Szegedy and G. Turán, *Threshold circuits of bounded depth*, J. Comput. System Sci. **46** (1993), no. 2, 129–154. doi:10.1016/0022-0000(93)90001-D

[HPV77] J. Hopcroft, W. Paul and L. Valiant, *On Time Versus Space*, J. ACM **24** (1977), no. 2, 332–337. doi:10.1145/322003.322015

[HS65] J. Hartmanis and R. E. Stearns, *On the computational complexity of algorithms*, Trans. Amer. Math. Soc. **117** (1965), 285–306. doi:10.1090/S0002-9947-1965-0170805-7

[IKW02] R. Impagliazzo, V. Kabanets and A. Wigderson, *In search of an easy witness: exponential time vs. probabilistic polynomial time*, J. Comput. System Sci. **65** (2002), no. 4, 672–694. doi:10.1016/S0022-0000(02)00024-7

[Ila26] R. Ilango, *SAT Reduces to the Minimum Circuit Size Problem with a Random Oracle*, SIAM J. Comput. **55** (2026), no. 3, FOCS23-309–FOCS23-359. doi:10.1137/24M1652568

[ILPS14] R. Impagliazzo, S. Lovett, R. Paturi and S. Schneider, *0-1 integer linear programming with a linear number of constraints*, preprint (2014), arXiv:1401.5512.

[ILW23] R. Ilango, J. Li and R. R. Williams, *Indistinguishability obfuscation, range avoidance, and bounded arithmetic*, in: Proceedings of the 55th Annual ACM Symposium on Theory of Computing (STOC 2023), 2023, pp. 1076–1089. doi:10.1145/3564246.3585187

[IM02] K. Iwama and H. Morizumi, *An explicit lower bound of* $5n - o(n)$ *for Boolean circuits*, in: Mathematical Foundations of Computer Science 2002, Lecture Notes in Comput. Sci. **2420**, Springer, Berlin, 2002, pp. 353–364. doi:10.1007/3-540-45687-2_29

[Imp95] R. Impagliazzo, *A personal view of average-case complexity*, in: Proceedings of the Tenth Annual Structure in Complexity Theory Conference, IEEE, 1995, pp. 134–147. doi:10.1109/SCT.1995.514853

[IP01] R. Impagliazzo and R. Paturi, *On the complexity of k-SAT*, J. Comput. System Sci. **62** (2001), no. 2, 367–375. doi:10.1006/jcss.2000.1727

[IPS13] R. Impagliazzo, R. Paturi and S. Schneider, *A Satisfiability Algorithm for Sparse Depth Two Threshold Circuits*, in: 54th Annual IEEE Symposium on Foundations of Computer Science (FOCS 2013), 2013, pp. 479–488. doi:10.1109/FOCS.2013.58

[IPS97] R. Impagliazzo, R. Paturi and M. E. Saks, *Size–Depth Tradeoffs for Threshold Circuits*, SIAM J. Comput. **26** (1997), no. 3, 693–707. doi:10.1137/S0097539792282965

[IPZ01] R. Impagliazzo, R. Paturi and F. Zane, *Which Problems Have Strongly Exponential Complexity?*, J. Comput. System Sci. **63** (2001), no. 4, 512–530. doi:10.1006/jcss.2001.1774

[IW97] R. Impagliazzo and A. Wigderson, $P = BPP$ *if* E *requires exponential circuits: derandomizing the XOR lemma*, in: Proceedings of the 29th Annual ACM Symposium on Theory of Computing (STOC 1997), 1997, pp. 220–229. doi:10.1145/258533.258590

[JMV18] H. Jahanjou, E. Miles and E. Viola, *Local reductions*, Inform. and Comput. **261** (2018), 281–295. doi:10.1016/j.ic.2018.02.009

[Juk10] S. Jukna, *Representing (0, 1)-matrices by boolean circuits*, Discrete Math. **310** (2010), no. 1, 184–187. doi:10.1016/j.disc.2009.07.011

[Juk12] S. Jukna, *Boolean Function Complexity*, Algorithms Combin. **27**, Springer, Heidelberg, 2012. doi:10.1007/978-3-642-24508-4

[Kan82] R. Kannan, *Circuit-size lower bounds and non-reducibility to sparse sets*, Inform. and Control **55** (1982), no. 1-3, 40–56. doi:10.1016/S0019-9958(82)90382-5

[Kar72] R. M. Karp, *Reducibility among Combinatorial Problems*, in: Complexity of Computer Computations, R. E. Miller and J. W. Thatcher (eds.), Plenum Press, New York, 1972, pp. 85–103. doi:10.1007/978-1-4684-2001-2_9

[KC00] V. Kabanets and J.-Y. Cai, *Circuit minimization problem*, in: Proceedings of the 32nd Annual ACM Symposium on Theory of Computing (STOC 2000), 2000, pp. 73–79. doi:10.1145/335305.335314

[KI04] V. Kabanets and R. Impagliazzo, *Derandomizing Polynomial Identity Tests Means Proving Circuit Lower Bounds*, Comput. Complexity **13** (2004), no. 1-2, 1–46. doi:10.1007/s00037-004-0182-6

[Kor21] O. Korten, *The hardest explicit construction*, in: 62nd Annual IEEE Symposium on Foundations of Computer Science (FOCS 2021), 2021, pp. 433–444. doi:10.1109/FOCS52979.2021.00051

[Kra19] J. Krajíček, *Proof Complexity*, Encyclopedia Math. Appl. **170**, Cambridge University Press, Cambridge, 2019. doi:10.1017/9781108242066

[KW16] D. M. Kane and R. Williams, *Super-linear gate and super-quadratic wire lower bounds for depth-two and depth-three threshold circuits*, in: Proceedings of the 48th Annual ACM Symposium on Theory of Computing (STOC 2016), 2016, pp. 633–643. doi:10.1145/2897518.2897636

[KW90] M. Karchmer and A. Wigderson, *Monotone Circuits for Connectivity Require Super-Logarithmic Depth*, SIAM J. Discrete Math. **3** (1990), no. 2, 255–265. doi:10.1137/0403021

[Lad75] R. E. Ladner, *On the Structure of Polynomial Time Reducibility*, J. ACM **22** (1975), no. 1, 155–171. doi:10.1145/321864.321877

[Lev73] L. A. Levin, *Universal sequential search problems*, Problemy Peredachi Informatsii **9** (1973), no. 3, 115–116 (in Russian); English translation in Problems Inform. Transmission **9** (1973), 265–266.

[Li24] Z. Li, *Symmetric Exponential Time Requires Near-Maximum Circuit Size: Simplified, Truly Uniform*, in: Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC 2024), 2024, pp. 2000–2007. doi:10.1145/3618260.3649615

[LST25] N. Limaye, S. Srinivasan and S. Tavenas, *Superpolynomial Lower Bounds Against Low-Depth Algebraic Circuits*, J. ACM **72** (2025), no. 4, 1–35. doi:10.1145/3734215

[LY22] J. Li and T. Yang, $3.1n - o(n)$ *circuit lower bounds for explicit functions*, in: Proceedings of the 54th Annual ACM Symposium on Theory of Computing (STOC 2022), 2022, pp. 1180–1193. doi:10.1145/3519935.3519976

[MS01] K. D. Mulmuley and M. Sohoni, *Geometric complexity theory I: an approach to the P vs. NP and related problems*, SIAM J. Comput. **31** (2001), no. 2, 496–526. doi:10.1137/S009753970038715X

[MS08] I. L. Markov and Y. Shi, *Simulating Quantum Computation by Contracting Tensor Networks*, SIAM J. Comput. **38** (2008), no. 3, 963–981. doi:10.1137/050644756

[Mul12] K. D. Mulmuley, *The GCT program toward the P vs. NP problem*, Commun. ACM **55** (2012), no. 6, 98–107. doi:10.1145/2184319.2184341

[Mur71] S. Muroga, *Threshold Logic and Its Applications*, Wiley-Interscience, New York, 1971.

[MW20] C. D. Murray and R. R. Williams, *Circuit lower bounds for nondeterministic quasi-polytime from a new easy witness lemma*, SIAM J. Comput. **49** (2020), no. 5, STOC18-300–STOC18-322. doi:10.1137/18M1195887

[NW94] N. Nisan and A. Wigderson, *Hardness vs randomness*, J. Comput. System Sci. **49** (1994), no. 2, 149–167. doi:10.1016/S0022-0000(05)80043-1

[OAI26] OpenAI, *The exponential time hypothesis is false*, OpenAI Math Release preprint, 17 October 2026, revised 25 October 2026, `https://github.com/openai/math/blob/main/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026/paper.pdf`.

[OS18] I. C. Oliveira and R. Santhanam, *Hardness Magnification for Natural Problems*, in: 59th Annual IEEE Symposium on Foundations of Computer Science (FOCS 2018), 2018, pp. 65–76. doi:10.1109/FOCS.2018.00016

[PPST83] W. J. Paul, N. Pippenger, E. Szemerédi and W. T. Trotter, *On determinism versus non-determinism and related problems*, in: 24th Annual Symposium on Foundations of Computer Science (FOCS 1983), IEEE, 1983, pp. 429–438. doi:10.1109/SFCS.1983.39

[PPSZ05] R. Paturi, P. Pudlák, M. E. Saks and F. Zane, *An improved exponential-time algorithm for k-SAT*, J. ACM **52** (2005), no. 3, 337–364. doi:10.1145/1066100.1066101

[Pud94] P. Pudlák, *Communication in bounded depth circuits*, Combinatorica **14** (1994), no. 2, 203–216. doi:10.1007/BF01215351

[Rag88] P. Raghavan, *Probabilistic construction of deterministic algorithms: Approximating packing integer programs*, J. Comput. System Sci. **37** (1988), no. 2, 130–143. doi:10.1016/0022-0000(88)90003-7

[Raz87] A. A. Razborov, *Lower bounds on the size of bounded depth circuits over a complete basis with logical addition*, Math. Notes Acad. Sci. USSR **41** (1987), no. 4, 333–338. doi:10.1007/BF01137685

[RM99] R. Raz and P. McKenzie, *Separation of the Monotone NC Hierarchy*, Combinatorica **19** (1999), no. 3, 403–435. doi:10.1007/s004930050062

[Ros08] B. Rossman, *On the constant-depth complexity of k-clique*, in: Proceedings of the 40th Annual ACM Symposium on Theory of Computing (STOC 2008), 2008, pp. 721–730. doi:10.1145/1374376.1374480

[RR97] A. A. Razborov and S. Rudich, *Natural Proofs*, J. Comput. System Sci. **55** (1997), no. 1, 24–35. doi:10.1006/jcss.1997.1494

[RSW22] H. Ren, R. Santhanam and Z. Wang, *On the range avoidance problem for circuits*, in: 63rd Annual IEEE Symposium on Foundations of Computer Science (FOCS 2022), 2022, pp. 640–650. doi:10.1109/FOCS54457.2022.00067

[RTS00] J. Radhakrishnan and A. Ta-Shma, *Bounds for Dispersers, Extractors, and Depth-Two Superconcentrators*, SIAM J. Discrete Math. **13** (2000), no. 1, 2–24. doi:10.1137/S0895480197329508

[San09] R. Santhanam, *Circuit Lower Bounds for Merlin–Arthur Classes*, SIAM J. Comput. **39** (2009), no. 3, 1038–1061. doi:10.1137/070702680

[San10] R. Santhanam, *Fighting Perebor: New and Improved Algorithms for Formula and QBF Satisfiability*, in: 51st Annual IEEE Symposium on Foundations of Computer Science (FOCS 2010), 2010, pp. 183–192. doi:10.1109/FOCS.2010.25

[Sav70] W. J. Savitch, *Relationships between nondeterministic and deterministic tape complexities*, J. Comput. System Sci. **4** (1970), no. 2, 177–192. doi:10.1016/S0022-0000(70)80006-X

[Sch02] U. Schöning, *A probabilistic algorithm for k-SAT based on limited local search and restart*, Algorithmica **32** (2002), no. 4, 615–623. doi:10.1007/s00453-001-0094-7

[Sch80] C. Schnorr, *A 3n-lower bound on the network complexity of Boolean functions*, Theoret. Comput. Sci. **10** (1980), no. 1, 83–92. doi:10.1016/0304-3975(80)90074-2

[SFM78] J. I. Seiferas, M. J. Fischer and A. R. Meyer, *Separating Nondeterministic Time Complexity Classes*, J. ACM **25** (1978), no. 1, 146–167. doi:10.1145/322047.322061

[Sha49] C. E. Shannon, *The Synthesis of Two-Terminal Switching Circuits*, Bell System Tech. J. **28** (1949), no. 1, 59–98. doi:10.1002/j.1538-7305.1949.tb03624.x

[Sip92] M. Sipser, *The history and status of the P versus NP question*, in: Proceedings of the 24th Annual ACM Symposium on Theory of Computing (STOC 1992), 1992, pp. 603–618. doi:10.1145/129712.129771

[Smo87] R. Smolensky, *Algebraic methods in the theory of lower bounds for Boolean circuit complexity*, in: Proceedings of the 19th Annual ACM Symposium on Theory of Computing (STOC 1987), 1987, pp. 77–82. doi:10.1145/28395.28404

[ST13] K. Seto and S. Tamaki, *A satisfiability algorithm and average-case hardness for formulas over the full binary basis*, Comput. Complexity **22** (2013), no. 2, 245–274. doi:10.1007/s00037-013-0067-7

[Tal14] A. Tal, *Shrinkage of de Morgan formulae by spectral techniques*, in: 55th Annual IEEE Symposium on Foundations of Computer Science (FOCS 2014), 2014, pp. 551–560. doi:10.1109/FOCS.2014.65

[Tar88] É. Tardos, *The gap between monotone and non-monotone circuit complexity is exponential*, Combinatorica **8** (1988), no. 1, 141–142. doi:10.1007/BF02122563

[Tod91] S. Toda, *PP is as hard as the polynomial-time hierarchy*, SIAM J. Comput. **20** (1991), no. 5, 865–877. doi:10.1137/0220053

[Val77] L. G. Valiant, *Graph-theoretic arguments in low-level complexity*, in: Mathematical Foundations of Computer Science 1977, Lecture Notes in Comput. Sci. **53**, Springer, Berlin, 1977, pp. 162–176. doi:10.1007/3-540-08353-7_135

[Val79a] L. G. Valiant, *Completeness classes in algebra*, in: Proceedings of the 11th Annual ACM Symposium on Theory of Computing (STOC 1979), 1979, pp. 249–261. doi:10.1145/800135.804419

[Val79b] L. Valiant, *The complexity of computing the permanent*, Theoret. Comput. Sci. **8** (1979), no. 2, 189–201. doi:10.1016/0304-3975(79)90044-6

[vzGG13] J. von zur Gathen and J. Gerhard, *Modern Computer Algebra*, third edition, Cambridge University Press, Cambridge, 2013. doi:10.1017/CBO9781139856065

[Wig19] A. Wigderson, *Mathematics and Computation: A Theory Revolutionizing Technology and Science*, Princeton University Press, Princeton, 2019.

[Wil05] R. Williams, *A new algorithm for optimal 2-constraint satisfaction and its implications*, Theoret. Comput. Sci. **348** (2005), no. 2-3, 357–365. doi:10.1016/j.tcs.2005.09.023

[Wil13] R. Williams, *Improving exhaustive search implies superpolynomial lower bounds*, SIAM J. Comput. **42** (2013), no. 3, 1218–1244. doi:10.1137/10080703X

[Wil14] R. Williams, *Nonuniform ACC circuit lower bounds*, J. ACM **61** (2014), no. 1, 1–32. doi:10.1145/2559903

[Wil14c] R. Williams, *Algorithms for Circuits and Circuits for Algorithms*, in: 29th IEEE Conference on Computational Complexity (CCC 2014), 2014, pp. 248–261. doi:10.1109/CCC.2014.33

[Wil16] R. R. Williams, *Natural Proofs versus Derandomization*, SIAM J. Comput. **45** (2016), no. 2, 497–529. doi:10.1137/130938219

[Wil18] R. R. Williams, *New algorithms and lower bounds for circuits with linear threshold gates*, Theory Comput. **14** (2018), Art. 17, 1–25. doi:10.4086/toc.2018.v014a017

[Wil26] R. Williams, *Simulating Time with Square-Root Space*, J. ACM **73** (2026), no. 3, 1–25. doi:10.1145/3798104

[Woe16] G. J. Woeginger, *The P-versus-NP page*, `https://wscor.win.tue.nl/woeginger/P-versus-NP.htm`, last updated 2016.

[WW26] M. Wu and M. Wang, *Circuit lower bounds for extended gate bases from subexponential counting*, preprint (2026), arXiv:2611.02560.

[Yan91] M. Yannakakis, *Expressing combinatorial optimization problems by linear programs*, J. Comput. System Sci. **43** (1991), no. 3, 441–466. doi:10.1016/0022-0000(91)90024-Y

[Yao85] A. C.-C. Yao, *Separating the polynomial-time hierarchy by oracles*, in: 26th Annual Symposium on Foundations of Computer Science (FOCS 1985), IEEE, 1985, pp. 1–10. doi:10.1109/SFCS.1985.49

[Zak83] S. Žák, *A Turing machine time hierarchy*, Theoret. Comput. Sci. **26** (1983), no. 3, 327–333. doi:10.1016/0304-3975(83)90015-4

[Zoo] S. Aaronson et al., *The Complexity Zoo*, `https://complexityzoo.net/Complexity_Zoo`.

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